

Leviathan-class Capital Assault Carrier

Build Specification, Revision H

Programme status: Single-consortium build. No competitive phase. **Supersedes:** Revisions A through G; Technology Request, all revisions.

1. Principal particulars

Type	Capital assault carrier, orbit-only
Overall diameter	3,356.9 m
Overall length, over drive fairing	3,536.9 m
Overall height, collar extended	730 m
Overall height, collar retracted	710 m
Hull thickness, hub	475 m
Hull thickness, outer band	~660 m, 630 m at the rim
Plating, dorsal	10 m, heaviest on the ship
Primary deck pitch	6 m
Ventral dish rim radius	1,130 m
Secondary core station and gallery radius	800 m
Dry mass, hull	~4.5 billion tonnes
Dry mass, nine cores	~300 million tonnes
Dry mass, mated	~4.8 billion tonnes
Hyperdrive, mated	Class 0.6
Hyperdrive, detached core	Class 0.6
Hyperdrive, in company	Class 2, the group's pace
Backup hyperdrive	Class 5
Sublight acceleration, mated	2,000 G
Sublight acceleration, cores deployed	~2,130 G
Sublight acceleration, detached core	500 G
Power	Hypermatter annihilation
Heat rejection	Through containment; batteries conventional
Endurance	Indefinite
Complement	~109,592

Embarked ground force	One Corps, 36,864 line
Embarked craft	1,188
Shield rings, hull	24
Cavity screen emitters	8
Sensor apertures, hull	96 plus cavity arrays
Umbilical booms	16
Station-keeping emitter pairs	16

Datum convention. The dorsal rim plane is $z = 0$. Positive is dorsal. The ventral rim plane sits at $z = -630$ m. The dorsal surface rises to +55 m across the hub, the fixed cap reaches +80 m, and the bridge collar crown stands at +100 m when extended. Radii are measured from the ship's vertical axis.

The mass figure is heavy by class standards and deliberately so. This is a monocoque with neutronium-impregnated plating over a doonium primary, built to absorb damage rather than avoid it, and to operate for decades without a yard. The hull works out at 0.93 t/m^3 against about 0.75 t/m^3 for comparable armored capital hulls, more than half of the difference in its dorsal plating (Section 20).

2. Hull form

A continuous monocoque disc. No ring, no spokes, no joints, no attachment interfaces anywhere in the primary structure.

2.1 Dorsal surface

A single continuous convex curve from rim to hub, rising 55 m across a 1,678 m radius: roughly 25.6 km radius of curvature. Visually subtle, and structurally inert. The hull's out-of-plane stiffness comes from its depth, a stiffened section 475 to 660 m deep working in bending, and against that depth a 55 m rise adds nothing of consequence at the scale of the whole disc. Nor does it at the scale of a panel: a skin panel 20 m between frames rises about 2 mm, against plating 10 m thick. Skin panels are sized as flat plate.

The curve's return is volume and sightlines. It adds roughly 244 million m^3 , all of it in the upper levels where berthing and the air group live, and it evens out the section: the hub goes from 420 m to 475 m against 630 m at the rim. It also tilts the outboard dorsal apertures outward (Section 7) and gives the bridge collar its horizon (Section 15).

Nothing penetrates this surface, and it carries the heaviest plating on the ship (Section 16). Batteries retract flush, sensors are conformal, and all nine core bowls are closed beneath it.

Fixed cap. Inboard of 250 m radius the surface continues to +80 m, covering the central core's crown at +66 m under the full dorsal grade. This is structure and does not retract.

2.2 Bridge collar

An annular structure at 250 to 450 m radius — 500 m inner, 900 m outer — extending 45 m above the local surface to a crown at +100 m. Sixteen viewport bays separated by structural piers, eight clocked to the web longitudes and eight falling between.

The collar retracts. It withdraws into an annular well with its floor at $z = -10$ m, after which iris shutters close flush and carry the dorsal curve across the well mouth. Retracted, the ship presents an unbroken curve from rim to rim with nothing standing proud anywhere.

Three constraints govern the geometry:

The ring sits outboard rather than at the hub because the central core's girth blocks a shallower well. The core is 204 m in radius at the datum and widens downward; at 150 m radius the well floor could only reach $z = +32$, leaving 23 m of depth for a 45 m collar. At 250 m radius the core does not reach the wall until $z = -40$, which gives 65 m of clear depth.

The collar carries no hull load. A retractable element cannot, which is why the shutters exist — they complete the structural surface independently. They also carry the full dorsal plating grade, so a stowed collar leaves no thin ring at a location every gunner knows.

Nothing crosses the centreline. The collar is a ring moving in a ring-shaped well, so the core's crown, the fixed cap and the flag stack all pass up the middle undisturbed. The central core can separate with the collar raised, stowed, or jammed at any intermediate position.

Doctrine. The collar is a manoeuvring and peacetime position. It stows before action, which removes the only aim point on the ship whose location every gunner knows.

2.3 Ventral surface

Concave. A single shallow dish spanning two-thirds of the disc's diameter, with the nine core bowls set into its inner surface and all sixteen launch mouths opening onto it. The crown falls gently from $z = -420$ over the hub to about $z = -470$ at 1,050 m radius. From there a knuckle turns the dish down to $z = -630$ at 1,130 m, and the turned-down edge forms a continuous structural ring. Outboard of the dish rim the ventral face runs flat at $z = -630$ out to the rim; that annulus carries the ventral shield rings and sensors (Sections 6 and 7).

Outer band. Everything outboard of the secondary core ring (Section 3), from about 1,050 m radius to the rim. It is deepest, about 660 m, at the dish rim.

Three reasons for the dish. It closes every hangar mouth and every core to the ship's plane, where other ships fight. The cavity is 210 m deep across a 2,260 m opening, so the cover is a band along the plane rather than a cone: the hub crown is masked within about 10.5° of the plane, and a Lance's south pole, 130 m above the rim plane, within 4° to 22° depending on bearing. Below that band about four-fifths of the lower hemisphere can see into the cavity; the orbital posture in Section 17.2 fills most of it with the planet, and the screen in Section 6.1 closes what the form cannot. A dished plate with a turned-down edge is far more rigid than a flat one of equal mass, and the knuckle ring stiffens the face that carries the sixteen mouths, the sixteen boom recesses and the eight screen emitters. And it creates a shielded mustering volume where craft form up before crossing the rim plane, which matters more on recovery than on launch.

The stiffness is directional, so the dish is sized against buckling rather than stress. Load arriving from the cavity meets the concave face and puts the dish in tension. Load from inside the hull, including the hull's own atmosphere, meets the convex face and puts it in compression.

2.4 Aft drive fairing

The hull's section grows continuously aft across roughly 90° of arc, between the web longitudes at 135° and 225° . Not a block, not an attachment. Frames and caps run through unbroken.

It stands aft of the rim circle on a cosine taper: 180 m at 180°, falling to nothing at both web longitudes, so the added plan form is a lune of about 0.315 million m². Depth is the full 630 m throughout and the dorsal surface runs flat at $z = 0$ across it under the full dorsal grade. Overall length is therefore 3,536.9 m against a diameter of 3,356.9, and the added section is about 198 million m³, which is carried in the density check of Section 20. Nothing changes from astern: the emitter face still spans the 2,374 m chord across the full section (Section 17.4).

2.5 Web longitudes

Eight radial lines at 0°, 45°, 90° and so on, clocked so one runs fore and one aft on a straight diametral line. These are the thickest continuous material in the hull and serve simultaneously as core key clocking, main vertical trunk routing, gallery spur routing, auxiliary power plant siting, shield ring clocking, and the primary radial load paths.

2.6 Atmospheric operation

None. The hull carries no entry provision of any kind: no ablative laminate, no refractory tiling, no stagnation-region rule. The concave ventral face would trap a standing shock and is unsuitable regardless. End-of-life disposal assumes a purpose-built repulsorlift cradle.

Entry protection appears only in the core specification, which is where it has always belonged.

2.7 Emplacement and immersion

The hull is orbit-only and carries no means of reaching a surface or leaving one. What follows is not a capability. It is recorded because the form has consequences that hold whether or not she is ever set down, and because a repulsorlift cradle can place her where she cannot lift out of.

Set into rock. At 0.96 t/m³ the hull is about a third the density of crustal rock, so an excavated void filled to the rim plane carries less afterwards than before: about 7.4 MPa against 16.7. Mean contact pressure over the 8.85 million m² footprint is 5.3 MPa at one standard gravity, 2.0 at Martian and 0.9 at lunar. Sound intact rock takes that; jointed or sedimentary rock creeps. Ground reaction arrives on the concave face of the dish, which is the sense Section 2.3 requires; the knuckle ring bears as a 550 m ring footing, and the central core bears directly at about 1.4 MPa.

Two conditions govern. The excavation must be uniform to several kilometres of depth, because the loaded volume under a footing 3.36 km across reaches that deep and differential stiffness beneath a disc this rigid is the failure case. And removing about 4.9 billion m³ of rock to put back a third of its mass unloads the site, which rebounds — of order a metre on a stiff mass at one standard gravity. Nothing in the cavity or the nine wells can separate while those voids are filled, so an emplaced hull is a fortress and not a carrier.

Afloat. Sealed in the Action state the hull is a closed monocoque with no penetrations, and it floats. With the ventral cavity flooded the mean density of the immersed form is 0.99 t/m³; with the cavity screen hardened (Section 6.1) it is 0.85. Gravity cancels out of flotation, so the figures hold on any world and only the medium decides.

Medium	Density	Draft, cavity flooded	Draft, screen hardened
Fresh water	1.000 t/m ³	622 m	538 m
Seawater	1.025 t/m ³	609 m	524 m
Dense brine	1.240 t/m ³	517 m	433 m

The rim stands clear by the balance of 630 m in each case, and the dorsal dome by 55 m above that at the hub, 80 m at the fixed cap. Hardening the screen closes about 740 million m³ — the cavity and the nine well annuli — and is worth 84 m of draft in any medium. Pressurising it adds nothing to buoyancy, since the boundary excludes water whether the volume holds gas or vacuum. What a fill at ambient pressure does buy is load path: the underside pressure, about 2.1×10^{13} N or some 45 percent of her weight, crosses the gas and arrives on the concave face of the dish rather than standing on the emitter ring. It costs roughly 45 million tonnes and five metres of draft, and the gas is drawn from a planetary atmosphere and vented back (Section 6.1).

Stability. Metacentric height is of order a kilometre against a 3.36 km beam and the roll period near a minute, which no sea can excite; the waterplane spans several wavelengths of the largest storm swell, so the disc averages a sea rather than following it. No stabilising system is fitted and no passive type will tune: a pendulum of one minute wants an arm of 894 m in a hull 630 m deep. The accelerations that do reach the decks — core release, a boom taking load, thrust — are step inputs, and are taken by inertial compensation, by the opposed-pair release rule and by the structural fields.

Submerged. Reserve buoyancy is shed by dropping screen segments in opposed pairs and flooding behind them; trim comes from the sixteen station-keeping emitters and from the nine wells. Hull-pressure operation is rated to 1,000 m, where external pressure is about 10 MPa, near the bedded contact case. The governing panel is the flat ventral annulus outboard of the knuckle: unsupported plate sized against weapon effect rather than against uniform external pressure. Below 1,000 m the cavity is pressurised and the screen is the pressure boundary, so the hull sees no differential; depth is then set by emitter output, and a boundary failure floods a volume built to flood.

A kilometre of water is about twelve times the areal density of the dorsal grade and it closes behind every shot, so it defeats beam weapons. It does not defeat anything that merely arrives, and it masks every battery, aperture and mouth she has. Immersion is a means of breaking contact, not a fighting state. Rejection is unaffected while containment holds (Section 5.5); a sector reduced to radiating warms the water around it and announces her position.

Cores. At 0.62 t/m³ the cores float on their own account. A Lance draws 244 m, a waterline at 9° N, which leaves every hangar mouth and the southern band dry and puts the entry face, the gear and the shield face under. Praetorian draws 404 m and stands 292 m clear. In brine at 1.24 the ratio is exactly one half and a core floats to its equator, waterline on the docking ring.

3. Core system

Nine detachable spherical cores: one central, 696 m in diameter, and eight secondary, 420 m in diameter. The eight secondaries, the Lances (Section 18.2), seat on a ring at 800 m radius, clocked between the web longitudes.

3.1 Mounting

All nine use one architecture, scaled twice.

Each core seats in a closed bowl conforming to its entire upper hemisphere and terminating at the equator. No aperture at the crown, no penetration of the dorsal surface. The bowl is a doubly-curved shell working in membrane action, and it stiffens the hull rather than weakening it.

Eight tapered longitudinal keys in each bowl wall engage matching channels in the core, clocked to the eight web longitudes. Taper brings all eight into simultaneous bearing on final seating, eliminating backlash under

load reversal. Service umbilicals — power, coolant, data, atmosphere — run within the key bodies and mate in the same motion.

Radial locking dogs at each bowl lip engage the core's equatorial box. The dogs carry restraint only.

Repulsorlift separation. Opposed emitter arrays on the bowl crown and the core's polar cap. Field effect, so no aperture, no mechanism, no consumable, no stored charge. Either surface's array alone can effect separation, so a dead core can still leave and a dead ship can still shed. Variable and reversible, making it the docking system as well as the release system.

Dogs never carry separation load. Repulsors never carry restraint load.

3.2 Vertical arrangement

	Station radius	Centre	Top	Bottom
Central core	0 m	$z = -282$	$z = +66$	$z = -630$, flush with rim
Secondary cores	800 m	$z = -290$	$z = -80$	$z = -500$

The central core's cap sits 130 m deeper into the cavity than the secondaries'. Nothing protrudes past the rim plane, making the disc a closed convex form in every direction — the best available shape for a shield envelope and free of snag geometry.

Below its bowl each core hangs in a cylindrical well. A secondary stands about 50 m proud of the crown at its station; the central core stands 210 m proud.

3.3 Core configuration

No exposed engine nozzles. All propulsion apertures are recessed behind two-part closures: an outer armored shutter flush with the plating, and an independent inner thermal seal deeper in the well. Iris or retracting-petal type throughout; hinged clamshells are prohibited.

North polar cap: main sublight drive cluster. Sealed under the closed bowl while seated, making ignition-while-mated a geometric impossibility rather than a procedural interlock.

Equatorial box: the heaviest structure on each core. On the central core it spans roughly $\pm 30^\circ$; on the secondaries it runs from 30° S to 40° N, to take the upper hangar deck at 35° N (Section 9.1). Four ring frames with shear webs between them. Carries the docking interface, the thruster wells, the hangar decks, the shield rings and the sensor apertures, and serves as the core's main frame.

Thruster belts: two, straddling the equator at roughly $\pm 8^\circ$, canted out of plane — upper belt north-of-plane, lower south-of-plane. Both belts together give pure translation; opposed gives a pure pitch or yaw couple; tangential pairs give roll. Full six-degree-of-freedom control with no dedicated attitude jets. Both belts are buried while seated. Thruster wells are clocked between hangar mouths and never share a longitude with one. On the central core, whose lower hangar deck lies at 7.6° N, that clocking is what keeps the upper belt clear of the deck, and it lets all nine cores keep one belt specification.

Hangar decks: two per core, northern hemisphere only. See Section 9.

Southern band, 30° to 50° S: landing pad bays, ventral ramps and the southern batteries, each behind the same two-part closures as every other opening. Gear legs reach 105 m at 30° S and 50 m at 50° S, which is what it takes to stand the core on its own pole clearance. On a Lance this band lies inside the well while seated, the well mouth falling at 49.6° S, so its closures face the well wall rather than the cavity, and its

batteries fire down through the widening gap between core and wall (Section 8).

South polar cap, south of 50° S: unbroken armored shield face and ventral repulsorlift array. Repulsorlift carries the landing load; gear takes stability only. Repulsorlift is a field projector requiring no aperture, which is why the shield face can be solid.

Powerplant: hypermatter, one per core.

Hyperdrive: one per core, Class 0.6. The Corps can therefore disperse across eight systems and reconverge.

Armament: every core is independently combat-capable, not merely armed for self-defence.

3.4 Central core role

Flag command, primary medical, and the ship's survivable volume. It lands as a relocatable fortress with its own shields, reactor, hyperdrive and repulsorlifts.

The endurance plant does not go with it. Agriculture, fabrication and stores stay in the hull. The core carries what the campaign needs; the hull carries what the ship needs.

Consequence to record. Primary medical lands with the core. From the moment Praetorian separates, the hull's medical capacity falls to departmental level, precisely when the ground campaign generates its heaviest casualties. Task force medical frigates hold that gap in orbit — see Section 18.3.

Launch while mated. Praetorian's two hangar decks, at 7.6° N and 18.8° N, tie into the galleries through the radial spurs, just as the Lances' decks tie into the gallery ring (Section 9.1). While mated she holds no independent launch authority: flag transport runs through the galleries and leaves by the sixteen launch mouths like everything else. Independent launch resumes on separation.

4. Propulsion

The drive is faired continuously into the hull across roughly 90° of aft arc and terminates in a segmented linear emitter spanning that arc. No discrete nozzles.

Distributed thrust is the point. A continuous band applies thrust uniformly along the whole arc, putting the aft hull into near-even hoop compression instead of local bending between concentrated loads. No thrust load travels circumferentially, because the nearest web longitude is never more than 22.5° away.

Minimum sixteen independently fed and throttled segments. Loss of any segment costs a fraction of thrust that trims out against the remainder, rather than a third of thrust and a large uncorrectable yaw moment.

Vectoring is electromagnetic, steering the exhaust particle stream across the emitter face. This gives true vectoring across a continuous band with a long lever arm at the outboard ends, and is the ship's primary yaw authority.

Hyperdrive. Class 0.6 mated. That is the rating Rothana's generator already achieves in the Acclamator, but no hull of this mass or form has carried it. The disc geometry is unfavourable for field generation, and the figure is achievable only because the consortium holds both halves of the problem (Section 19). In company she keeps her group's pace; Class 0.6 is held for movement alone (Section 18.5).

5. Power generation

Hypermatter annihilation throughout, hull and cores alike. A hypermatter plant's constraints are heat rejection and containment, and both are answered by siting.

No hypermatter plant in the hull, main or auxiliary, is sited inboard of any hangar volume. A craft that reaches a hangar, a gallery spur or a launch mouth shall not find a reactor within reach; that is how the reference vessel was lost (Section 8).

5.1 Main plant

Four reactors aft, at 150°, 165°, 195° and 210°, all four in the faired volume outboard of the hull's rear face (Section 2.4). They stand at the quarter points of the emitter arc, each feeding four of its sixteen segments directly and cross-tying to the main trunks on the 180° spine. None sits on a web longitude: a spine already carries its trunk, its auxiliary plant, its growing blocks and three shield generator halls, and will not take a reactor hall as well.

Four rather than one because the linear emitter is already segmented sixteen ways and wants distributed feed, and because losing a quarter of generation is survivable where losing all of it is not.

They sit aft and outboard because that places the ship's full radial depth plus an armored bulkhead between a containment failure and everything that matters. Twenty metres of armor stands between each hall and the next, and between every hall and the ordnance blocks inboard of it; the halls cross-tie in pairs, so a stern hit costs a quarter of generation rather than half. The tighter cluster is the price of putting all four where the fairing has depth to give them.

5.2 Distributed auxiliaries

Eight auxiliary plants, one per web longitude at about 1,400 m radius, on the same spine that carries the main power trunks. That is outboard of the dish rim and of every hangar volume, roughly midway between the dorsal and rim shield rings each plant feeds, and beside the growing decks it keeps alive.

Each is sized to run its own 45° sector unaided, every load at once: one shield ring in each of the three bands, its share of the batteries, its sensors, and the life-support and agricultural load in that sector. Eight independent sectors that happen to be interconnected, rather than one grid with eight branches. The aft sector carries no agricultural load, its outer band being propulsion and ordnance (Section 14); the plants are identical regardless and are sized to the seven that do.

Battle damage therefore takes out a sector, not a system. Losing one auxiliary thins the shield across its 45° sector in all three bands and costs the growing decks in that wedge, and the neighbouring sectors close over both.

5.3 The base load rule

Life support, recycling, agriculture and fabrication are unsheddable. They are constant loads that never throttle down and shall not be sacrificed to combat demand.

This is the requirement that makes indefinite endurance real. A vessel that starves its growing decks to hold its shields has traded ten years of endurance for one engagement. The distributed architecture enforces the rule by priority and sizing rather than by separate supplies. Base load has first call on every auxiliary. Each auxiliary carries its sector's base load and its full combat load at the same time, so no sector ever has to

choose between them. The main plant reinforces combat loads through the trunks and never draws on base-load capacity. When demand or rejection is short the order of shedding is fixed: the cavity screen first, then propulsion above station-keeping, then the batteries, then the shields. Base load does not appear in the order.

5.4 Core plants

Seated cores tie into the sector grid at their own longitudes and contribute rather than draw. Nine reactors idling in the hull is real capacity, so the ship is better powered mated than deployed — the correct direction, since mated is when it carries the most people and the most load.

Bidirectional charging. Any core can accept a full charge from the ship and any core can feed the ship. A dead core recharges from the grid; a dead ship recovers from nine independent plants sitting in its own wells. Given that the cores are also the crew survival volume, this redundancy is worth its switchgear.

5.5 Thermal rejection

Hypermatter annihilation rejects its waste heat through its own containment rather than through radiating surface. The products are already partly outside normal spacetime and the field that holds them carries the heat out with them, so rejection scales with containment throughput and not with hull area. That is what makes a sealed monocoque of this displacement possible with no radiator vanes, no exhaust ports and no thermal aperture anywhere on the vessel.

The boundary is the rule. A load that runs inside a containment or field boundary rejects through it. A load that discharges through a boundary cannot, and is handled conventionally.

Load	Rejection
Main plant, auxiliaries and core plants	Through containment
Endurance plant, growing decks, fabrication, habitability	Through containment, by way of the sector grid
Hyperdrive, shield rings, cavity screen when hardened	Through containment
Propulsion at cruising and manoeuvring thrust	Through containment
Sublight emitter at full thrust	Conventional at the throat; duty-cycle limited
Turbolaser and point-defence emplacements	Conventional; mass sink and cycle (Section 8)
Cavity screen when attenuating	Partly open; treated as conventional

The batteries are the exception by construction. A turbolaser rejects at the throat, outboard of any containment, at the instant the bore is open to vacuum. Dallorian liners, the mass sink behind each emplacement and cycle time therefore set the sustained rate of fire, not available power (Section 8). The emitter discharges the same way but across sixteen segments of a continuous band, so full thrust is duty-cycle limited and everything below it is not.

Degraded rejection. Field rejection requires a field. A sector that loses containment reverts to radiating through its plating, and the dorsal face is the emergency radiator: 8.86 million m², the only surface with a clear view of the sky in either attitude, good for something of order ten gigawatts. That is orders below plant output, so a sector in that state sheds in the order set by Section 5.3 and its neighbours carry what they can.

6. Shields

Twenty-four rings on the hull, in three bands, all clocked to the web longitudes.

Band	Count	Siting	Coverage
Rim	8	One per web longitude	135° of the plane, centred on its longitude
Dorsal	8	Two-thirds radius	Upper hemisphere, 135° of azimuth
Ventral	8	Outer band, outside the dish rim	Lower hemisphere, 135° of azimuth; bridges the wells

Each ring projects across 135° of azimuth centred on its own longitude: its own 45° sector and both neighbours'. Three overlapping rings therefore cover any bearing in each band. Lose one and its two neighbours still cover its sector, at reduced strength. Lose an entire band and the other two still enclose the ship, thinner overhead or underneath.

Clocking matters more than the count. The web longitudes are the thickest continuous material in the hull and already carry the main trunks, so every generator sits on a spine that already has the feed. The rim rings hold the only face with a clear field through the full 360° of the plane, and the dorsal rings sit behind the heaviest plating: the two faces where the envelope takes the most punishment.

The envelope bridges the nine wells when cores are deployed. The closed convex hull form means it is a simple lens rather than anything with a hole through it.

Each core carries four rings at 90° spacing on its equatorial box — thirty-six across the nine, independent of the hull's envelope and surviving separation.

6.1 Cavity screen

Twenty-four rings enclose the hull. A twenty-fifth surface closes the cavity, and it exists because the dish's geometry cannot: the form closes the cavity to the plane and leaves about four-fifths of the lower hemisphere looking in (Section 2.3).

Siting. Eight emitters on the knuckle ring between 1,050 and 1,130 m radius, clocked to the Lance longitudes from 22.5°, which puts them between the launch mouth pairs and clear of the boom roots. Clocking them off the web longitudes is deliberate. A spine already carries its ventral ring, its trunk and its auxiliary feed; a screen segment on the same longitude would lose both surfaces to one hit, and offset it does not. Generators sit in the outer band inboard of the ventral rings, fed from both neighbouring spines.

Surface. About 4.01 million m² spanned, near enough half the dorsal face, whether drawn flat or domed. Depth is adjustable across the knuckle's span. Flat at the rim plane is the rest position and, with cores seated, the limit: Praetorian's south cap is flush with that plane, so the hardened surface sits 10 m below it. Bulged outward it stands furthest from the hull and holds the most working volume, at the cost of the closed convex form the rest of the ship is built around. Drawn inboard against the dish it is the strongest membrane for a given feed and the shortest projection distance, which makes it the deployed-state setting, where it closes nine open wells and sixteen mouths at once. Standoff from a seated Lance's polar cap is 130 m.

Off. The cavity is open and held by battery alone, as in Section 8.

Attenuating. Graded transmission. Craft, booms and descending cores cross at low speed with no gating, and the surface takes energy weapons and high-velocity fire. A membrane that passes a landing craft

passes a starfighter flying slowly, which is how the reference vessel died, so cavity defence is not reduced by one emplacement on this account.

Hardened. No transmission. Transit is by gated aperture: locally suppressed patches, opposed pairs only, never two adjacent segments, never more than a quarter of the surface at once. Core release becomes four events of two apertures each, which is the release sequence Section 11.1 already fixes.

Supply. The screen is a main plant load drawn through the trunks rather than a sector load, and it stands first in the shed order (Section 5.3). An isolated sector gives up its screen segment and keeps its dish, its ventral ring and its guns.

The cavity is not pressurised in vacuum. The screen is a barrier, not a vessel, and 740 million m³ at one atmosphere is about 900,000 tonnes of air — more than the recycling plant could replace after a single flicker. Pressurisation is an immersion case only, where the gas comes from outside the ship (Section 2.7).

7. Sensors

A disc is its own horizon. Dorsal apertures are blind below the ship's plane, because 630 m of doonium is in the way. Coverage therefore requires three bands, matching the shield and battery split, so that sensors and fire control share siting and aperture-to-battery latency is near zero.

Band	Count	Siting
Dorsal	32	Sixteen each at one-third and two-thirds radius
Rim	24	One every 11.25° across the forward 270°
Fairing	8	Four on each face of the drive fairing, forward of the emitter, raked aft
Ventral	24	Outer band, outside the dish rim
Hub cap	8	On the fixed cap, highest point in either collar state

The aft 90° of the rim is emitter face (Section 4). The rim band therefore stops at the forward 270°, and the eight fairing apertures take the stern arc: the same division the rim battery makes with the aft ordnance cells (Section 8).

Ninety-six on the hull, plus close-in arrays inside the ventral cavity. Those cavity arrays serve traffic control and terminal defence for the sixteen mouths and nine cores; they are not search sensors. No search sensor is sited inside the dish, where it would see only the far wall of its own cavity.

Eight per core, on the equatorial box.

The dorsal curve contributes here. On a flat plate every aperture shares a normal, giving redundancy without added coverage. On a convex shell the outboard apertures tilt outward, so the same count covers a wider solid angle. Modest at 25 km curvature, but free.

No masts, towers, or external antennae anywhere on the vessel. All apertures are conformal.

8. Armament

The reference hull's documented failure was that only a fraction of its armament could bear on a single target: its equatorial batteries covered too little of the hull. The disc form fixes this only if the guns are sited where the geometry pays.

Primary batteries — rim. Continuous heavy turbolaser emplacements around the forward 270° of the rim, about 7,910 m of the 10,546 m circumference; the aft 90° is drive emitter (Section 4). The rim is the only surface with a clear field through the full 360° of the ship's plane and offers the longest lever arm from the centre. Half the rim bears on any target in the plane. Within 45° of the bow all of that half is battery; on the beam, three-quarters of it; dead astern, half of it, which is a quarter of the rim. There the aft ordnance cells make up the difference.

Secondary batteries — both faces. Medium turbolasers in concentric bands on both faces: dorsal at one-third and two-thirds radius, ventral at one-third and in the outer band, which stands further out because the knuckle is fully claimed (Annex C). The disc's own body masks the far face, so each side delivers near-hemispherical coverage rather than compromising a single battery across both.

Point defence — distributed, weighted ventral. A single small craft reaching the hull is how the reference vessel died. Density is highest around the nine bowls and the sixteen launch mouths, which are the only openings that exist.

Cavity defence — dedicated interior layer. Anything crossing the rim plane is inside the outer defences and simultaneously adjacent to sixteen hangar mouths and nine cores. This is the most important point-defence installation on the vessel. Emplacements are sited for coverage rather than arc avoidance; deconfliction with friendly craft is handled by fixed flight corridors through the cavity.

It is sized against the operating attitude, not against the general case. In campaign posture the ship holds ventral-down over the objective, low enough that the planet fills most of what can see into the cavity (Section 17.2). Ground-based anti-orbital fire and surface-launched interceptors arrive on exactly the bearing where every opening is. Cavity defence is therefore specified at a density above the ventral face band rather than below it, and it is the one battery on the ship that must engage upward-climbing targets at close range as its primary case. See Section 17.2.

Ordnance — aft segments. Missile and torpedo cells between the drive fairing and the two aft core bowls, already reactor-adjacent and already the most heavily protected interior volume on the ship. They are the ship's answer to the stern arc, where the rim battery bears only from its flanks.

Core batteries. Each secondary core mounts batteries in its southern band (Section 3.3), contributing to ventral firepower while mated by firing down through the gap between core and well wall, and bearing through full arcs once detached. The central core, sitting deeper in the cavity, contributes primarily to cavity defence while mated and fights independently once landed.

Retraction. All emplacements stow flush behind plating when not engaged.

Cooling. The batteries are the one conventional thermal load on the ship (Section 5.5). Sustained rate of fire is set by the Dallorian throat liners, by the mass sink behind each emplacement and by cycle time, not by available power. The geometry already carries half the answer: only half the rim bears on any target in the plane, so half the rim battery is always cooling, and an emplacement stowed flush behind plating is an emplacement in its cycle.

9. Air group and hangar system

1,188 craft, split by role rather than by location.

Type	Quantity	Based
Superiority and multirole starfighters	576	Hull
Strike and bomber craft	288	Hull
Assault gunships, troop-carrying	144	Lances, 18 each
Heavy-lift gunships, vehicle-carrying	48	Lances, 6 each
Heavy landing craft	48	Lances, 6 each
Utility, shuttle, medical evacuation	48	Lances, 6 each
Medical evacuation, flag flight	18	Praetorian
Command transport and utility, flag flight	18	Praetorian

The fleet element never deploys, so the ship is not defenceless the moment it drops its cores. The organic element goes down with the task force at 36 craft per core, Praetorian included. Her flight is weighted to casualty evacuation because primary medical lands with her (Section 3.4).

Those 36 are not the delivery mechanism. The core lands and the troops walk out the ventral ramps (Section 3.3). Embarked craft are post-landing mobility and close support.

9.1 Annular galleries

Two continuous ring corridors run through the hull at 800 m radius, one above the other, and pass through all eight secondary cores. Radial spurs on the eight web longitudes carry both decks inward to the central core and outward to the launch mouths. Together they tie every core to every other core and to every launch mouth without leaving pressurised volume. Elevator trunks inside each core connect the two decks. Each gallery deck is 80 m wide and 24 m in the clear — four primary bays — with handling alcoves outboard and the radial spurs on the same section. Through a Lance the gallery runs a 406 m chord at 15° N and a 344 m chord at 35° N.

The gallery decks are ship structure and are fixed. Core hangar latitudes are derived from them, not the reverse.

Deck	Height above rim plane	Absolute	Secondary core	Central core
Lower gallery	394 m	$z = -236$	15° N	7.6° N
Upper gallery	460 m	$z = -170$	35° N	18.8° N

Both decks sit above every core's equator. This is forced by the bowl geometry: only the region above the equator conforms to the core, so a hangar mouth below it would face a widening gap and could not mate.

The constraint produces the right answer anyway. Every hangar mouth on every core ends up in the northern hemisphere, and the southern hemisphere — entry face, shield, repulsorlift array, landing gear —

carries no hangar mouths at all.

When cores deploy, each gallery breaks into eight arcs behind pressure doors. Every arc retains its own launch mouths, and its own spur to the central core while she remains seated.

9.2 Launch mouths

Sixteen, two per inter-aperture segment, opening onto the ventral cavity at approximately two-thirds radius, in the inner face of the dish's knuckle. Connected to the lower gallery by the outward spurs and lifts. Each mouth is 120 m wide and 60 m high in the clear, the pair in a segment on 240 m centres, with approach corridor and lift trunk on the same section. The binding case is the 64 m Gozanti of Section 13.5, which passes with margin. Mouth centres fall at $r = 1,118$ and $z = -545$, which is where two-thirds radius meets the knuckle profile rather than a figure imposed on it (Annex B).

Ventral siting rather than rim or dorsal. The rim carries the primary battery across the widest weapon arc on the ship and should not be cut. Dorsal siting would cut the heaviest plating on the ship and split traffic across two hemispheres. Everything that leaves this vessel goes down — cores, craft and cargo alike — so the ship presents its unbroken dorsal surface to the enemy while it works.

10. Complement

Human-crewed throughout. No droid control dependency in any command, fire control, or ground-force chain. When the reference vessel was destroyed over Naboo, that dependency took its army down with it; the single point of failure does not exist in this architecture.

Composition. Mixed establishment. Clone naval personnel throughout the technical branches; Republic Navy commissioned officers in command and staff appointments. The commanding officer is a Republic Navy captain; the executive officer is a clone naval commander. This division is deliberate — it places fleet command authority in the commissioned chain while the ship's technical continuity rests with clone personnel who will serve the hull for its operational life.

Element	Personnel
Ship's company	42,500
Secondary core crews, $8 \times 1,000$	8,000
Air group, aircrew and maintenance	9,280
Embarked ground force	49,812
Total	~109,592

Ship's company: command and staff 500, navigation and flight operations 1,500, gunnery 8,000, engineering 10,000, damage control 4,000, hull and structural maintenance 6,000, medical 2,000, supply 4,000, sensors and electronic warfare 1,500, security 2,000, central core detachment 3,000.

11. Embarked ground force

Grand Army establishment, base-four throughout. Squad 9, platoon 36, company 144, battalion 576, regiment 2,304, legion 9,216, corps 36,864.

Line strength: one Corps — four legions, 36,864 troopers. Four hulls lift an Army; sixteen lift a Systems Army.

Element	Personnel
Line infantry, four legions	36,864
Armor and artillery crews	6,048
Combat engineers	1,700
Ground medical	1,300
Logistics and transport	2,000
Ground air liaison and forward control	400
Civil administration and occupation staff	1,500
Total	49,812

The armor and artillery figure covers dedicated crews for 288 AT-TE at seven — a pilot, a spotter and five gunners — 96 SPHA at thirty, and 576 Saber-class tanks at two, a pilot and a gunner. A Jedi officer may take a Saber into action alone with an astromech; the establishment does not count on it. AT-RT, BARC and Juggernaut positions are filled from line and support strength, since all three are ridden or driven by the units they carry.

The civil administration element does not scale with line strength. Administering a captured population is sized to that population.

11.1 Distribution

36,864 divided by eight cores is 4,608 exactly — two regiments. Any two cores together land a full legion.

Cores release in opposed pairs to keep the centre of mass on axis. **Each opposed pair is therefore one legion, and four releases put the entire Corps on the ground at eight separate sites with no aircraft involved.**

Berthing. The Corps berths aboard its cores, not in the hull. Line infantry, armor and artillery crews, combat engineers, logistics and transport, and ground air liaison berth in the Lances with their vehicles, alongside each Lance's crew of 1,000. Ground medical and civil administration berth in Praetorian with primary medical and the flag. The hull stages vehicles and stores for them (Section 14).

11.2 Vehicles

Type	Quantity
Heavy assault walkers, six-legged class	288
Self-propelled heavy artillery	96
Medium armor, Saber-class repulsor tanks	576
Heavy armored troop transports	192
Light scout and reconnaissance walkers	1,152
Speeder bikes	3,072
Engineering, bridging, mobile fabrication	384
Support, utility and cargo	~10,000

Heavy artillery is the compact walker pattern, under 60 m long, carried twelve to a Lance. Longer patterns cannot be stowed and landed through a Lance's ventral ramps at that count.

11.3 Occupation stores

Prefabricated garrison base modules for four legions in permanent installation. Field hospital capacity. Water purification and independent power generation sized to a planetary population centre. Civil administration and communications infrastructure. Humanitarian ration reserve sized to the occupied population, not the embarked force.

12. Endurance

Indefinite. Decomposed into five independent problems, each solved separately.

Atmosphere and water. Closed-loop recycling at 99.5% recovery, with makeup drawn from any icy body or planetary source. Physicochemical plant alone falls short of that figure; the growing decks close the remainder, which is one more reason they are unsheddable (Section 5.3).

Subsistence. Onboard agriculture sized to feed the whole task force rather than the flag alone, because the flag replenishes the group (Section 18.4). The allowance is 45 m² of growing area per person: about 8.25 million m² for the task force's ~183,300, carried on roughly 516,000 m² of footprint at sixteen tiers, under a tenth of the outer band. The aft sector carries none of it, so the other seven carry about 73,600 m² of footprint each, and the two flanking the aft block carry theirs on the bow side (Annex B). It assumes high-intensity lighting and a staple mix of grains, legumes, tubers and oil crops. At that allowance the decks close the atmosphere loop with margin; calories are the binding requirement. Polyculture mandatory; a single blight shall not be capable of ending a deployment.

Fuel. Hypermatter bunkering at decade scale under operational duty cycle.

Spares and munitions. Onboard fabrication from raw feedstock, plus in-situ resource acquisition and refining. A vessel that must return to a yard for a structural spare has a defined endurance, not an indefinite one.

Crew. The binding limit is human, not material. Berthing standard, rotation accommodation, medical capacity for 109,592, and habitability for deployments measured in years. At this scale the question ceases to be habitability and becomes demography.

All four material limits are underwritten by the base load rule in Section 5.3.

13. Replenishment and station-keeping

The flag's endurance is indefinite. Every escort's is finite. This section is the mechanism by which the first extends to the second, and it is the reason a Leviathan group requires no fleet train.

13.1 No cradles

Fitted berths were considered and rejected. A cradle sized for a ship that is present occasionally is a permanent void in the structure and a permanent gap in the plating, returning nothing for every hour it stands empty. It also cannot serve two sizes: a fitting sized for a 1,137 m Venator is empty air around a 325 m Arquiten's, and a fitting sized for the Arquiten's cannot take the Venator at all.

Nothing on this vessel exists to accommodate something that is not there.

13.2 Umbilical booms

Sixteen booms, two per inter-aperture segment, stowed in recesses on the cavity's inner surface. Each recess closes with the same two-part armored shutter used on every thruster well, hangar mouth and collar well aboard: an outer shutter flush with the plating, an inner seal deeper in the well.

Deployed, a boom extends down and outward to a hull holding station beneath the cavity. Hypermatter, ordnance, power, data and atmosphere pass through a single connection. Stowed, there is nothing. The cavity surface is unbroken plating between the launch mouths.

Boom roots are sited on the web longitudes. A boom under load is a cantilever pulling on the cavity surface, and that surface is a shell working in membrane action. Rooting each boom on the thickest continuous material in the hull puts the load into the spine that already carries the trunks, the auxiliary plants and the shield rings, rather than into shell that would have to bend to take it. The fuel and power feeds are correspondingly short.

Two per segment so that each of the eight gallery arcs retains its own replenishment capability after core release, and so that one boom may be down for repair without costing that segment its function.

13.3 Station-keeping emitters

Paired tractor and repulsor emitters at each boom root — sixteen pairs.

Neither effect alone can hold a position. A tractor beam only pulls, so it corrects drift away from the flag and does nothing about drift toward it. Run opposed, the pair has authority in both directions along the same axis. This is the same push-pull arrangement that governs the thruster belts on every core: two effects that each do half the job, arranged so the halves complete each other.

The field holds the receiving ship; the receiving ship idles. Without it, an escort holds station on its own drives while a boom reaches for it — hardest precisely when it matters most, for a damaged hull with degraded control arriving to reload. The correct allocation is the reverse. The flag has 4.8 billion tonnes of reference mass, twelve powerplants, and nothing else to do during a transfer.

Two things follow. The boom no longer requires a compliant tip chasing a drifting target and may be a rigid connection made to a fixed one, which is simpler and carries more. And station-keeping tolerance ceases to be a matter of metres of slop.

Emitters are recessed behind the same armored shutters as the booms.

13.4 Emitter sizing — the recovery case

Sizing is set by the hardest case, not the routine one. Holding a cooperative Arquitens is a small load. Arresting a tumbling capital hull is not.

Stated requirement. The vessel shall be able to take an uncontrolled hull of up to Venator mass — no drives, no power, no control — arrest its tumble, walk it to station beneath the cavity, and hold it there while a boom supplies power, atmosphere and damage-control feed.

This is a recovery capability the task force does not otherwise possess and it costs nothing beyond emitters already installed. It is written as a requirement rather than left to emerge, because it is what determines emitter output.

13.5 What comes inside and what does not

Gozanti-class cruisers, approximately 64 m: internal. Eight berth in the lower gallery, one per inter-aperture segment, handling heavy freight movement within the group. They pass through openings that already exist and require no modification. Counted as fleet auxiliaries, not against the air group's 1,188.

Everything larger stays outside. Arquitens and Pelta at roughly 300 m, Acclamator II and Victory I, and Venator at 1,137 m all hold station beneath the cavity and are serviced by boom. None enters the hull.

A hull holding station beneath the cavity is below the rim plane, where the dish gives it no cover. The flag's body screens most of its upper hemisphere and the Screen Group Venators stationed below the flag cover its lower (Section 18.3), so the protection is retained without pretending a capital hull can dock.

13.6 Traffic separation

Booms are sited between launch mouths and share no corridor with them. Replenishment and flight operations run independently through the cavity, so a transfer in progress does not close a segment's sortie capability. When the screen is hardened, booms transit it under the same gate discipline as craft, on their own apertures (Section 6.1).

14. Interior arrangement

The organising rule is depth. This hull is a plate, and the safest place in a plate is its mid-plane.

Hub, mid-plane — the citadel. Combat bridge, primary reactor control, fabrication plant, munitions magazines. Deepest material in every direction. Permacite bulkheads, turadium blast doors in series.

Outer band, mid-plane and below, forward 270° — the endurance plant. Agriculture, recycling, decade-scale stores. The largest single claim on the ship, sited where volume is cheapest and damage least likely to reach. It stops at $r = 1,500$: the outer 178 m of the hull is a weapons band over the battery rows, and carries no stores. The extra depth in the outer band falls on the ventral side, so the plant's centre of volume sits below mid-plane, near the fabrication plant and far from the dorsal batteries.

Upper third — air group and berthing. Aircraft handling, and quarters for the 51,780 ship's company and aircrew. Berthing sits high because it must reach both the galleries and the bridge, and because the dorsal surface carries no openings and no traffic. The dorsal curve's added volume lands here. The Corps and the Lance crews berth aboard their cores (Section 11.1).

Mid-depth at 800 m radius — the galleries. The two gallery decks straddle the local mid-plane, which lies at $z = -203$ at that radius: the upper deck 33 m above it and the lower 33 m below. Everything moving between hull, cores and launch mouths travels on them (Section 9.1).

Lower third — ventral traffic. Launch mouth approaches, vehicle handling, stores staging, bowl rims. Vehicles and stores stage here and load into the cores through the gallery connections.

Aft 90° — propulsion and ordnance. Reactor halls in the fairing at 150°, 165°, 195° and 210°, ordnance cells inboard of them, armored bulkheads between. No endurance plant stands in this arc. The 180° spine keeps a corridor of $\pm 7.5^\circ$ through the ordnance blocks for its trunk, its auxiliary and the two mouth lift trunks.

Web longitudes — the spines. Eight main trunks running full depth, each carrying lifts, power, coolant and damage-control routing, with an auxiliary powerplant at about 1,400 m radius and shield rings in all three bands.

Hub — the flag stack. Bridge collar, flag plot beneath it, central core command deck below that, passing up through the middle of the collar ring.

15. Command and control

Annular bridge collar, retractable, sixteen viewport bays with structural piers between them. See Section 2.2 for geometry and retraction.

A disc has no bow to look over. Threat arrives from any bearing, and a linear bridge facing one direction answers a question this hull does not ask. The collar is also the only position on the ship with a clear horizon in all directions, since the dorsal surface falls away from it at about four degrees to the rim.

Combat bridge, deep in the citadel behind armored bulkheads. Identical stations, no viewports. The ship fights from below and commands from above; losing the collar costs visibility, not the vessel.

Flag stack. Bridge collar, flag plot at the fixed cap's base, central core command deck immediately below in the hull. When the core drops, the flag element steps down one level and takes its headquarters with it. The bridge it leaves behind keeps the ship.

No masts, towers, or external antennae anywhere. Deleting the dorsal communications array is what formally makes this a battleship rather than a control ship, and removes the reference hull's defining vulnerability.

16. Materials

Selected per load case rather than from a single palette.

Application	Material
Hull primary structure	Doonium caps and frames, ferrocarbon lattice webs
Hull plating	Neutronium-impregnated heavy quadanium, graded by exposure: heaviest by far on the dorsal surface (grades below)
Reactor-adjacent underlayer	Agrinium
Bowl shells	Alusteel over ferrocarbon framing

Application	Material
Keys, locking dogs, latch hardware, landing struts	Duranium
Bowl bearing pads	Dallorian alloy
Reactor containment	Turadium with neuranium shielding, agrinium underlayer
Emitter throat liners	Dallorian alloy
Bridge collar structure and well shutters	Alusteel, quadanium facing; well shutters at full dorsal grade
Core shell	Alusteel primary, ferrocabon framing, doonium at concentrated load paths
Core equatorial box	Doonium rings and shear webs
Core ventral shield	Quadanium substrate, refractory ceramic facing, ablative laminate at stagnation region
Citadel bulkheads	Permacite
Blast doors	Turadium
Viewports	Armored transparisteel behind quadanium shutters

Plating is graded by exposure. The dorsal face is the heaviest on the ship by a wide margin, two and a half times any other face, because it is the face turned toward the threat in both working attitudes: toward the surface on a dorsal-down approach and toward space in campaign posture (Sections 17.2 and 17.3).

Surface	Grade	Exposure
Dorsal surface, fixed cap, collar-well shutters	10 m	Toward the surface on approach; toward space in campaign posture
Rim	4 m	Edge-on to the orbital plane in either attitude
Ventral face, cavity and core wells	4 m	Toward the surface after turnover

At a working plating density of 8 t/m³, plating totals about 1.2 billion tonnes, a quarter of the hull, and the dorsal face carries 0.71 billion of it.

Entry protection appears only in the core specification. The hull carries none.

Anti-boarding is a geometry problem, not a materials problem. No conventionally available material resists a lightsaber. Defence is by chokepoint count, approach length, and doors in series.

17. Operating states and orbital doctrine

17.1 Configuration states

Cruising. Nine cores seated, bowls closed, bridge collar extended, dorsal curve broken only by the collar. Air operations through sixteen launch mouths onto the cavity. Nine core reactors contributing to the sector grid. Cavity screen attenuating.

Action. Bridge collar stowed, shutters flush, batteries deployed from their recesses. The ship presents a closed convex form with no protrusion but the mounts themselves, which stand a maximum of 12 m proud at full stroke, and the shield rings at 8 m when energised. No tower, no mast, no aim point at the hub. Overall height drops to 710 m. Cavity screen hardened; transit by gated aperture.

Deployed. Cores released in opposed pairs, one legion per pair. The hull retains 864 fighters and bombers, all rim and face batteries, and the full endurance plant. Mass drops by roughly 300 million tonnes, about six percent, so sublight acceleration rises to about 2,130 G. Agility improves by only a few percent, since the Lances sit at 800 m radius, well inboard of the rim that dominates the ship's inertia. The shield envelope bridges the nine open wells. Galleries break into eight arcs, each retaining its own launch mouths. Generation falls to the four main and eight auxiliary plants. The cavity screen draws inboard against the dish and closes the open wells and the mouths behind the envelope.

Independent core operations. Every core carries a hypermatter plant and a Class 0.6 hyperdrive, so the Corps can disperse to eight systems and reconverge. The central core operates as a mobile fleet headquarters or a landed fortress.

17.2 Orbital attitude

The vessel has a fighting orientation, not merely an operating one. In campaign posture over an objective she holds ventral face toward the planet and dorsal face toward space.

Everything that must reach the surface already faces it: nine cores, sixteen launch mouths, the whole cavity. Nothing reorients, nothing crosses the hull, and the dish rim closes every opening to the horizon.

Everything that must survive faces the threat: unbroken dorsal plating, eight shield rings, thirty-two sensor apertures.

The rim serves both at once. It is edge-on to the orbital plane in either attitude, so the primary batteries and eight further shield rings cover the horizon regardless of which way she is turned.

The weak bearing is beneath her. About four-fifths of the lower hemisphere can see into the cavity (Section 2.3), and in campaign posture the planet fills most of that, so ground fire arrives exactly where the openings are. Three requirements follow: cavity defence is sized against this case rather than the general one (Section 8); the screen is held hardened except during release and recovery (Section 6.1); and cores are put down early rather than loitered with wells full.

Campaign posture is flown low. The planet fills the cavity's exposure only from low altitude. From 300 km over an Earth-sized world the limb sits about 17° below the horizontal, and the planet covers most of what can see in. A ring of open space just above the limb still sees into the cavity at grazing angles, and it widens with altitude; the Screen Group Venators stationed below the flag hold it (Section 18.3).

The attitude is fixed while she works. She cannot roll to present fresh plating or unmask a different arc without turning her openings toward whatever she is fighting. Damage accumulates on one face, and damage control is planned accordingly.

17.3 Arrival and turnover

She arrives dorsal-down and turns over only when the orbit is hers.

On approach there is nothing to gain from facing the planet. Cores cannot drop into a contested orbit, fighters should not launch into unsuppressed anti-orbital fire, and the cavity is the one thing she cannot afford to expose. Dorsal-down places unbroken armor between the surface and everything soft.

She fights in that attitude without penalty. Only the two face-mounted battery bands trade places; the rim never cares.

Turnover is a decision, not a phase. It is executed after anti-orbital fire is suppressed, not during suppression, and as a single continuous rotation. Half-over presents the cavity edge-on to both the surface and to open space simultaneously, which is the worst geometry the ship can occupy. If fire reopens mid-rotation the correct response is to complete the turn, never to reverse it.

Launch mouths are masked through part of the manoeuvre, so the ship's own fighters cannot cycle. Escort fighter cover is required for those minutes (Section 18.3). The screen is hardened throughout the rotation: half-over is the one interval in which the cavity faces the surface and open space at once, and it is also the one in which nothing needs to cross it (Section 6.1).

17.4 Stern aspect

From directly astern the emitter face spans a 2,374 m chord across the hull's full 630 m section. In the plane the dish rim hides the cavity from every bearing, so what is particular to the stern is what a pursuer sees instead: the drive face, backed by the most heavily protected interior volume on the ship, flanked by rim plating seen almost edge-on. It is the correct aspect to present when disengaging. Only the flanks of the rim battery bear on this arc; the aft ordnance cells carry the rest (Section 8).

17.5 Multi-hull stacking

Two or more hulls over the same objective stack at separate altitudes rather than sharing one. The lower hull's dorsal face is covered by the upper hull's cavity defence, and the upper hull's rim covers the lower hull's approach, so a vertical column has no weak bearing between its members.

Four hulls lift an Army and, in column, hold every arc from the surface to high orbit.

18. Class establishment and task organization

18.1 Hulls

Sixteen, in four divisions of four. A division lifts one Army. The class complete lifts a Systems Army.

Hull names alternate between great beasts and great works, the class being named for the first of them.

Division	Hulls
First	Leviathan (lead), Bastion, Behemoth, Acropolis
Second	Colossus, Barbican, Basilisk, Ziggurat
Third	Titan, Redoubt, Kraken, Monolith
Fourth	Goliath, Ravelin, Hydra, Cromlech

18.2 Core designations

Class convention, identical across all sixteen hulls. The flag core is Praetorian. The eight secondaries are Lance One through Lance Eight, numbered by aperture clocking from 22.5° through 337.5°.

Opposed pairs are One/Five, Two/Six, Three/Seven and Four/Eight. Each pair lands one legion.

18.3 Task force

Each hull serves as flag of a task force of forty-seven hulls, plus eight freighters berthed internally.

Element	Class	Hulls	Crew per hull	Role
Flag	Leviathan	1	109,592	Corps lift, orbital dominance, sustained presence
Bombardment Group	Acclamator II	8	700	Anti-orbital suppression, one per landing site
Screen Group	Venator	4	7,400	Orbital screen, fighter cover, ventral cone coverage
Atmospheric Group	Victory I	4	5,200	Capital-scale close support inside the atmosphere
Picket and Escort	Arquitens	18	750	Distant picket, close screen, core escort
Support	Pelta	4	900	Medical and supply
Courier	Consular, Charger c70	8	73	Liaison, dispatch, Jedi transport
Freight	Gozanti	8	8	Internal, berthed in the lower gallery

Task force personnel: approximately 183,300, counted as ships' companies. Escorts embark no ground troops; the Corps is the task force's whole ground force. **Starfighters:** 2,640 — the flag's 864 fighters and bombers, 420 per Venator and 24 per Victory I. The flag's 324 core-based gunships, landing craft, utility and flag-flight craft are additional.

The counts follow from the flag's own architecture rather than preference:

Eight Acclamator II because there are eight cores. Each works its assigned landing site before the corresponding Lance releases, shortening the interval during which the flag loiters with full wells and an exposed cavity.

Eighteen Arquitens because each core carries an independent hyperdrive. Two per core, Praetorian included, so no core makes an independent jump unescorted; a core jumping independently keeps its escorts' pace (Section 18.5).

Four Venator, two stationed below the flag. A Venator lower in the well looks up into the same cone the cavity opens onto, covering the bearing the flag cannot. The remaining two station above. All four hold the air through turnover.

Four Pelta because primary medical lands with Praetorian (Section 3.4).

Four Victory I because the flag is orbit-only and a landed core is a fortress rather than a gunship. One per legion, providing the only capital-scale fire support available in the atmosphere. They are underpowered at sublight and set the group's pace in transit (Section 18.5).

Eight couriers because the Corps can disperse across eight systems. One per Lance for dispatch and liaison, with Jedi transport as tasked.

18.4 Escort endurance

The flag's endurance is indefinite; every escort's is finite. She replenishes the group from her own foundry, growing decks and recycling plant through the sixteen umbilical booms of Section 13, and the growing decks are sized for the whole group (Section 12).

Escort endurance therefore becomes the flag's endurance. The group's reach is set by how long the escorts are willing to stay, not by a fleet train. This inverts the usual relationship, in which a formation's reach is limited by its slowest tender.

18.5 Movement in company

Endurance flows from the flag to the escorts; speed flows the other way. She keeps station on her group rather than running ahead of it. In company the group jumps at the rating of its slowest hulls, Class 2 for the Arquitens, Peltas and couriers, and transits at sublight at the pace of the Victory I. A core on an independent jump keeps pace with its two Arquitens the same way, which is what makes the escort rule in Section 18.3 hold on arrival, and a Lance's courier, also Class 2, keeps pace with both. The Gozantis ride inside at whatever the flag makes.

Class 0.6 is held in reserve for movement alone, when the flag or a core must leave faster than its escorts can follow.

19. Consortium responsibilities

Element	Lead
Prime, hull structure, slipway	Kuat Drive Yards
Hyperdrive field geometry, non-convex hull solution	Hoersch-Kessel Drive
Hyperdrive generators, core systems, ground complement	Rothana Heavy Engineering
Closed ecology, endurance plant, damage control	Mon Calamari yards
Air group and hangar systems	Kuat Systems Engineering

The mated hyperdrive rating is the one specification no single member could have met. Hoersch-Kessel's Lucrehulk is the proven case of a hyperdrive field wrapped around a non-convex hull, and Hoersch-Kessel has never had a fast generator to put behind it: the Lucrehulk is Class 2. Rothana builds the Class 0.6 generator proven in the Acclamator and has only ever installed it in clean convex hulls. The consortium exists because those two facts are complementary.

20. Open items

None remain.

Dry mass is closed. The dorsal regrade adds about 0.57 billion tonnes, bringing the hull to about 4.47 billion tonnes, carried as ~4.5 in Section 1. Checked by density, that is 0.93 t/m³ over about 4.83 billion m³ enclosed, the drive fairing's 198 million m³ now dimensioned and included (Section 2.4), and the cores' 300 million tonnes over 487 million m³ gives 0.62 t/m³. Comparable armored capital hulls run near 0.75 t/m³. The hull sits about a quarter above that, more than half of the excess in its dorsal plating; the cores sit below it, as cores built around troop and vehicle volume should. The figures stand.

Sublight acceleration is a requirement rather than a result, so the emitter is sized to hold 2,000 G at the heavier mass.

Shield ring count and siting, sensor fit, power architecture, plating grade, crew composition, replenishment method, orbital doctrine, task force movement, class establishment and dry mass are closed as of this revision.

Annex A — Source and continuity notes

Author's reference, outside the in-universe specification. Figures come from the canon and Legends verification of Revision B; where the continuities disagree, the figure used is given with the alternative.

A.1 Task force (Section 18.3)

- Venator: 1,137 m is the Legends length; canon says only that it exceeds 1,000 m. The 420-starfighter complement is the canon figure (192 V-19 or V-wing, 192 Eta-2, 36 ARC-170); a Legends RPG loadout gives about 192. Crew 7,400; hyperdrive Class 1.
- Acclamator II: crew 700, the Incredible Cross-Sections figure for the Acclamator, applied to the II; some RPG stat blocks give 4,280. Hyperdrive Class 0.6 per Attack of the Clones: Incredible Cross-Sections; Starships of the Galaxy (2007) gives 0.75.
- Victory I: crew 4,798 plus 402 gunners; Class 1; Legends describes its ion engines as underpowered.
- Arquitens: 325 m. Hyperdrive Class 2 with a Class 12 backup, the documented rating for the Imperial command-cruiser rework, applied to the Republic light cruiser, for which no sourced rating was found; fan stat blocks give Class 1. Its crew is the weakest figure in the table. The specification uses 750; fan RPG stat blocks run as low as 100, which would bring the task force total to about 171,600.
- Pelta: 282 m; crew about 900; Class 2.
- Couriers: read as Consular-class cruisers in the Charger c70 retrofit, crew 73 (reference figure; the unarmed Consular carries 9); Class 2.
- Gozanti: 63.8 m; crew taken as 8; sources range from 4 to 14.

A.2 Ground force (Section 11)

- AT-TE: crew of seven — pilot, spotter, five gunners.
- TX-130 Saber: standard crew of two, pilot and gunner. The solo Jedi-and-astromech configuration appears on screen but is not the establishment figure.
- SPHA-T: crew of thirty; the StarWars.com Databank gives no fewer than thirty clone troopers per platform. Reference sources disagree on length; figures of about 34 m and 140.2 m both appear. The specification uses the compact scale.
- Grand Army establishment: squad 9 through corps 36,864 follows the Legends structure, in which the 9,216 unit is a legion (also given as brigade). Current canon uses “legion” loosely, as with the 501st. The Army (147,456) and Systems Army (589,824) tiers are less consistently documented and serve here as the class's planning convention.

A.3 Hyperdrive and sublight (Sections 1, 4, 17, 18.5)

- Class 0.6: the Acclamator is the precedent. No source gives a sub-Class-1 rating for a hull remotely this massive, or for 420–696 m spheres, so the rating is an engineering claim resting on the consortium rationale.
- Sublight benchmarks are Incredible Cross-Sections and Legends figures: Imperial-class about 2,300 G, Executor about 1,230 G. 2,000 G mated sits within that scaling.

- Movement in company (Section 18.5) follows US carrier strike group practice with the Nimitz and Ford classes: the carrier holds the group's transit speed rather than its own maximum.

A.4 Reference hull (Sections 8, 10, 15, 19)

- The Lucrehulk is about 3,170 m in Legends; canon says over three kilometres. Hyperdrive Class 2. Sources document the limited hull coverage of its equatorial batteries and its loss at Naboo to a single starfighter reaching the reactor. Revision B's "roughly a quarter" was not found in sources, so the specification says "only a fraction."

A.5 Consortium (Section 19)

- Rothana and Kuat developed the Acclamator jointly, the precedent for a multi-builder capital ship. Revision B's "only proven solution" and "fastest generator of its generation" were design lore; the specification restates both on the sourced records.

A.6 Materials (Section 16)

- All twelve named materials appear in sources, and the assignments fit their described properties. Ferrocabon is attested as a structural material in Coruscant construction rather than in starships, so its use for hull lattice webs is an extrapolation.

A.7 Plating (Section 16)

- No source gives a density or standard thicknesses for quadanium plating. 8 t/m³ is armor-steel density, used as a working figure for mass.
- The 4 m rim and ventral grade sits at battleship proportion: belt armor on the Iowa and Yamato classes ran about 0.11–0.16% of hull length, which at this diameter is 4–5 m. The dorsal grade is two and a half times that.
- The fairing's crown adds about 0.315 million m² of dorsal surface at full grade, some 0.03 billion tonnes, which sits inside the figures carried in Section 16.
- Revision B's thin dorsal grade is taken as 2 m, so the regrade adds 8 m across about 8.86 million m², about 0.57 billion tonnes. The fixed cap rises by the same 8 m because it had only 6 m of depth over the central core's crown.

A.8 Engineering basis

- Shallow shell (Section 2.1): rise-to-span is about 1:61. Membrane action matters for thin shells; in a section hundreds of metres deep, bending of the deep section governs. Locally the curve is flatter still: a 20 m panel on a 25.6 km radius rises about 2 mm, and snap-through needs a rise comparable to the plate's thickness. The curve is non-structural at every scale.
- Dish (Section 2.3): dished heads outperform flat heads of equal thickness. A head loaded on its convex side is governed by buckling, as in ASME rules for formed heads with pressure on the convex side.
- Dish cover (Sections 2.3, 17.2): the rim masks a point in the cavity from any bearing shallower than the arctangent of its height above the rim plane over its horizontal distance to the rim edge on that bearing. The hub crown, 210 m up and 1,130 m from the rim, is masked within 10.5°. A Lance's south pole, 130 m up at 800 m radius, is masked within 3.9° looking across the dish, 9.3° sideways and 21.5° outward, which leaves it visible from about 81% of the lower hemisphere. From altitude h over a world of radius R the limb sits $\arccos(R/(R + h))$ below the horizontal: about 17° at 300 km over an Earth-sized world.
- Flotation (Section 2.7): displacement is the solid volume below the rim plane, about 4.87 billion m³ with the cavity flooded and 5.62 billion with it closed, against a waterplane of 8.85 million m². Gravity cancels, so the drafts hold on any world; dense brine is taken at 1.24, the Dead Sea figure. Liquid methane at about 0.45 does not float her.

- Bearing (Section 2.7): contact pressure is weight over plan area. Intact granite and anorthosite run 100–250 MPa unconfined with allowable bearing an order below that; jointed and sedimentary rock does not qualify. Rebound follows from the hull being about a third the density of the rock it displaces.
- Hull pressure (Section 2.7): 1,000 m of seawater is about 10 MPa. The limit is stated rather than derived, and is set by the flat ventral annulus, which is unsupported plate sized against weapon effect.
- Growing area (Section 12): 40–50 m² per person supplies full diet and atmosphere closure under high-intensity lighting (NASA baseline values; Wheeler et al., *Frontiers in Astronomy and Space Sciences*, 2021). About 20–25 m² covers oxygen alone. Biosphere 2's food and oxygen shortfall is the cautionary case. The specification uses 45.
- Water recovery (Section 12): the ISS life-support system demonstrated 98% total water recovery in 2023. 99.5% exceeds any real system and is carried as a bioregenerative figure.
- Density (Sections 1, 20): the Imperial I-class, listed at 40 million tons over 1,600 m, implies about 0.75 t/m³ on a volume estimate of about 5.4×10^7 m³. Floating steel warships run about 0.1–0.35 t/m³ on enclosed volume, submarines about 1.0.
- Hull volume method (Section 20): a cylinder to $z = -630$ plus the dorsal cap, less the ventral cavity (crown to $z = -470$ at 1,050 m, knuckle to $z = -630$ at 1,130 m), the nine core wells and the collar well. The drive fairing is dimensioned in Section 2.4 and included: a cosine lune outside the rim circle, 180 m deep at 180° and nothing at the web longitudes, 0.315 million m² in plan over the full 630 m section, about 198 million m³. That is what takes the enclosed volume from 4.63 to 4.83 billion m³ and the density from 0.96 to 0.93. The elliptical knuckle of Annex B takes a further 25 million m³ out of the hull against the straight-cone approximation used here, which is inside this figure's rounding.

A.9 Thermal rejection (Section 5.5)

- Sources give thermal hardware but no thermal budget: heat exchangers and radiator surfaces on Imperial-class hulls, reactor coolant systems throughout, and a thermal exhaust vent as the documented failure of the first Death Star. No source gives a reactor output, a radiator area or an operating temperature, so no heat balance can be closed from the record.
- Hypermatter is described as existing partly outside normal spacetime. Rejection through containment is an extrapolation from that description rather than a documented mechanism, in the same class as ferrocabon in A.6. It is stated in Section 5.5 rather than left implicit, because at this displacement the alternative is a hull that glows.
- Gun cooling is the one thermal limit the sources do honour: capital turbolasers are described as recharge- and cooling-limited, with crews managing heat between shots. Section 8 follows that, and the exception falls out of the boundary rule rather than being asserted.
- Radiative capacity of the dorsal face, used for the degraded case: about 7.5 GW at 350 K and 30 GW at 500 K over 8.86 million m² against a 3 K sky.

Annex B — Modeling datum set

Author's reference, outside the in-universe specification. This annex fixes the dimensions a physical or digital model needs and the body does not state. Where a figure appears in the body it is not repeated unless a model needs it in one place. Every figure originating here is a decision of this revision, taken to be consistent with what the body already fixes; none of it is drawn from source material. Geometry only — finish, livery, markings and interior styling are not specified.

B.1 Frame and conventions

- Origin on the ship's axis at the dorsal rim plane, $z = 0$, positive dorsal, as Section 1 states. Radii r from the axis. Longitude θ measured from the bow at 0°, increasing to starboard, so the stern is 180° and the web longitudes

fall at 0°, 45°, 90° and so on. Lance stations fall at $22.5^\circ + 45n$, Lance One at 22.5° .

- Deck levels are numbered n from the lower gallery deck: $z = -236 + 6n$. Negative n is ventral. The convention is used throughout this annex.
- All surface figures are to the moulded surface, the outside of plating. Grades in Section 16 lie inboard of it. Structural depths given here are inboard of the moulded surface unless stated.
- Model tolerance ± 0.5 m. Nothing in the specification is fixed closer than 0.1 m, and the two figures quoted to that precision — the 3,356.9 m diameter and the 1,678.45 m radius derived from it — are the only ones that warrant it.

B.2 Hull surfaces

- The hull is a solid of revolution everywhere except the aft 90°, the collar well and the nine bowls. Seven of the eight inter-aperture segments are identical; the two aft segments carry the reactors, the ordnance cells and the emitter. A model needs one 45° sector, the two aft segments, the collar, and the two core types.

Surface	Definition
Dorsal surface	Spherical cap, radius of curvature 25,638 m, rising 55 m from the rim at $z = 0$ to the axis. Surface slope 3.75° at the rim, zero on the axis.
Fixed cap	Spherical dome inboard of $r = 250$, radius of curvature 1,206 m, reaching +80 on the axis. It meets the dorsal curve in a knuckle at $r = 250$, on the collar well's inner wall; the two are not tangent.
Collar well	Annulus $r = 250$ to 450, vertical walls, floor at $z = -10$. Depth 64 m below the local surface. Sixteen iris leaves close flush and carry the dorsal curve across the mouth.
Ventral crown	Paraboloid: $z = -420 - 50 (r \div 1,050)^2$, from the axis to $r = 1,050$, $z = -470$. This is the profile that reproduces both the 50 m proud figure of Section 3.2 and the enclosed volume of Section 20.
Knuckle	Quarter-ellipse, semi-axes 80 m radial and 160 m vertical, horizontal tangent at the crown and vertical where it meets the ventral rim plane at $r = 1,130$, turning into the flat annulus in a right angle — the structural ring of Section 2.3.
Ventral annulus	Flat, $z = -630$, from $r = 1,130$ to the rim. The bearing face of Section 2.7 and the governing panel under external pressure.
Rim	Cylinder, $r = 1,678.45$, $z = 0$ to -630 , over the forward 270° .
Drive fairing	Outer boundary $r = 1,678.45 + 180 \cos 2(\theta - 180^\circ)$, for $135^\circ \leq \theta \leq 225^\circ$. Full depth, dorsal surface flat at $z = 0$ across it. Maximum reach $r = 1,858.45$ at the stern.
Bowls	Spherical, conforming to each core's upper hemisphere, terminating at the equator. Running clearance 2 m, shell 12 m, so the bowl's moulded radius is the core radius plus 14 m.
Core wells	Cylindrical below each bowl lip, radius the core radius plus 6 m, running to the ventral crown at that station.

B.3 Deck grid

- Primary pitch 6 m: 4.5 m clear, 1.5 m structure and services. The grid is anchored on the lower gallery at $z = -236$ rather than on the rim plane, because that is the only anchor that puts both galleries on it — the upper gallery at $z = -170$ falls exactly eleven levels above.
- Sub-modules: 3 m half-levels for berthing mezzanines, offices and growing tiers; 12 m doubles for vehicle decks, workshops and magazines; 24 m quads for handling bays and the galleries. Every module is a whole multiple of the primary pitch, so nothing floats off the grid.
- Level counts: at the rim the stack runs $n = -65$ ($z = -626$, above the ventral grade) to $n = 37$ ($z = -14$, under the dorsal grade), 103 levels. At the hub it runs $n = -30$ ($z = -416$, above the crown) to $n = 46$ ($z = +40$), 77 levels. The collar and fixed cap stand above the grid and are not numbered.
- Gross deck area at pitch is about 805 million m^2 in the hull and 81 million in the nine cores, before deducting trunks, shafts, machinery volumes, magazines and tankage. Roughly four-fifths is deckable floor; the balance is volume that is never decked.

Feature	Level	z
Highest hub level	$n = 46$	+40, under the cap
Flag plot	$n = 42$ to 46	+16 to +40, annular, $r = 210$ to 250
Highest rim level	$n = 37$	-14, under the dorsal grade
Upper gallery	$n = 11$	-170
Lower gallery	$n = 0$	-236
Citadel mid-plane	$n = -9$	-290
Growing decks	$n = -25$ to -18	-386 to -344, sixteen tiers at 3 m
Hub floor	$n = -30$	-416, above the crown
Launch mouth sill	$n = -57$	-578
Lowest rim level	$n = -65$	-626, above the ventral grade

B.4 Framing and structure

- Ring frames every 20 m of radius, 84 from the axis to the rim, continuous through the aft block and around the bowls. Radial frames: 64 primaries on 5.625° , doubling at $r = 200, 400, 800$ and 1,600, giving 1,024 at the rim and holding every skin panel between 10 and 20 m on a side — the panel size Section 2.1 sizes as flat plate.
- Deck plating 0.3 m, on girders 1.5 m deep set on the 20 m frame lines. Trunk and shaft walls 3 m. Pressure bulkheads at every gallery arc boundary, 6 m, with 6×6 m doors. Citadel bulkheads 12 m of permacite; blast doors 4 m of turadium, 20 m wide by 12 m high, in series of three at every citadel approach.
- Plating grades from Section 16 lie inboard of the moulded surface: 10 m dorsal, including across the fairing crown and the collar-well shutters; 4 m at the rim, on the ventral annulus, through the cavity and in the core wells.

B.5 Spines, trunks and auxiliaries

- Each web longitude carries a main trunk 60 m wide by 60 m deep running the full section, containing four lift shafts of 12 m, the power and coolant mains, and damage-control routing. Trunks are continuous from $n = 37$ to $n = -65$ and are the only full-depth penetrations of the deck stack. Each stands at $r = 1,230$ to 1,290, threaded between the dorsal generator hall inboard of it and the auxiliary block outboard.

- Each auxiliary plant occupies a machinery block 200 m by 200 m in plan and 300 m deep, centred at $r = 1,400$ and $z = -300$ on its spine, with a containment sphere of 80 m at its heart and an agrinium underlayer to its bulkheads. It sits outboard of every hangar volume, as Section 5 requires, and inboard of the ventral shield ring on the same spine. The sector's two growing blocks flank it tangentially (B.11).
- Sector boundaries fall on the Lance longitudes, not the spines, so each auxiliary sits at the middle of the 45° sector it feeds.

B.6 Cores

- Both core types use one internal grid, anchored on the core's own equator plane and matching the hull's 6 m pitch, which is what lets a gallery deck run straight through a seated core without a step.

Feature	Lance	Praetorian
Diameter	420 m	696 m
Station, centre	$r = 800$, $z = -290$	$r = 0$, $z = -282$
Equatorial box	30° S to 40° N, 40 m deep	$\pm 30^\circ$, 60 m deep
Ring frames in box	Four, with shear webs	Four, with shear webs
Keys	Eight, 30 m wide, 12 m proud, taper 1:40	Eight, 50 m wide, 20 m proud, taper 1:40
Locking dogs	Sixteen at the bowl lip	Twenty-four at the bowl lip
Hangar decks	15° N and 35° N, through-corridors 60×20 m, two mouths each	7.6° N and 18.8° N, 60×20 m, eight mouths each on the spur longitudes
Thruster belts	$\pm 8^\circ$, 32 wells per belt, 12 m	$\pm 8^\circ$, 48 wells per belt, 18 m
Drive cluster	Above 60° N, eight nozzles of 30 m	Above 60° N, twelve nozzles of 45 m
Reactor	Containment sphere 60 m, at the centre	Containment sphere 100 m, at the centre
Landing gear bays	Eight, 30×20 m, 30° to 50° S	Twelve, 45×30 m, 30° to 50° S
Ventral ramps	Four, 40 m wide by 15 m, centred 40° S, clocked between gear bays	Six, 50 m wide by 18 m, centred 40° S
Shield rings	Four at 90° on the box	Four at 90° on the box
Sensor apertures	Eight on the box, 12×6 m	Eight on the box, 16×8 m
Command deck	None; local control off the upper hangar deck	Flag command deck at $z = +28$, the highest full level in the core

- Internal arrangement, both types: drive cluster and its plant in the northern cap; berthing between the two hangar decks; vehicle decks from the equator down to 30° S; stores, water and the ventral ramp lobbies below that; the southern band and polar cap as Section 3.3 sets them. A Lance berths about 6,880 — 4,608 line, its share of the ground support elements and its crew of 1,000 — at roughly 60 m^2 per person of habitable claim.

B.7 Galleries, spurs, mouths and booms

- Galleries 80 m wide by 24 m clear at $r = 800$, two decks, eleven levels apart. Radial spurs on the eight web longitudes, 60 m by 24 m, inward to Praetorian and outward to the mouths, with a lift trunk of 40 m by 24 m at each mouth head.
- Launch mouths 120 m wide by 60 m high in the clear, cut normal to the knuckle: sill at $r = 1,125$, $z = -575$, head at $r = 1,106$, $z = -515$, centre at $r = 1,118$, $z = -545$. Two per segment on 240 m centres, which places them at $\theta = 45n \pm 6.15^\circ$. The lift trunk over each stands at $r = 1,106$ to $1,130$, cut to the knuckle where it meets it rather than carried square off the mouth head.
- Boom recesses on the web longitudes between each mouth pair, two per segment stacked radially, both roots on the knuckle curve: $r = 1,108$, $z = -520$ and $r = 1,128$, $z = -596$. Each recess is 60 m by 40 m behind a two-part shutter, and the lower root stands high enough that its recess clears the ventral grade at $z = -626$. A boom is a lattice cantilever 40 m square in section, 400 m stowed, articulated at the root, with a reach of 900 m below the rim plane. Stowage is tangential in the outer band at $r = 1,130$ to $1,170$: the upper boom lies toward the bow at $z = -540$ to -500 , the lower toward the stern at $z = -618$ to -578 .
- Station-keeping emitter pairs at each boom root: a tractor and a repulsor aperture of 20 m on 30 m centres, behind the same shutter as the boom.
- Gozanti berths in the lower gallery, one per segment: a bay 80 m by 30 m off the outboard side, with the ship struck down through the mouth lift.

B.8 Batteries, apertures and emitters

Installation	Datum
Heavy rim battery	Mounts 30 m across on a 40 m pitch, three rows at $z = -120, -315$ and -510 , over the forward 270° : 197 to a row, 591 in all, the arc taking a whole number of pitches and no more. Recess depth one diameter behind the shutter; barbette 44 m; feed trunk 15 m over 90 m to a ready magazine of 40 by 40 by 30 m, with a thermal sink block of 30 by 30 by 20 m behind each mount (Section 5.5).
Medium face batteries	Mounts 15 m on a 30 m pitch, barbette 22 m, 45 m deep: 117 in each one-third band at $r = 559.5$, 234 in the dorsal two-thirds band at $r = 1,119$, and 278 in the ventral outer band at $r = 1,331$. The ventral band stands well outboard of two-thirds radius because the knuckle is fully claimed by the mouths, the screen emitters and the boom recesses, and the stowed booms sweep out to $r = 1,236$.
Point defence	Mounts 8 m on a 120 m grid across both faces and the rim, thickened to a 40 m grid within 200 m of any opening: about 1,280 in all. Barbette 12 m, 20 m deep.
Cavity defence	640 mounts, forty a segment, on the crown, the knuckle inner face and the nine well lips, sited for coverage rather than arc avoidance (Section 8).
Ordnance cells	Tubes 6 m by 24 m in two blocks between $r = 1,050$ and 1,400 on levels $n = -40$ to -10 , from 146° to 172.5° and from 187.5° to 214° . The $\pm 7.5^\circ$ corridor they leave on the 180° spine carries that spine's trunk, its auxiliary and the two mouth lift trunks. Inboard of the reactor halls behind an armored bulkhead.
Sensor apertures	Conformal panels 20×8 m on the dorsal, rim, fairing and ventral groups; 12×6 m on the hub cap. Cavity arrays are 6 m domes, three a segment, 48 in all.
Shield ring emitters	Each a 60 m toroidal aperture flush in the plating, standing 8 m proud when energised; ventral emitters at $r = 1,560$. Generator halls on the spine, set in the gaps between battery rows: rim 140 m at $r = 1,500$ to 1,674, $z = -288$ to -148 ; ventral 140 m at $r = 1,500$ to 1,630, $z = -483$ to -343 ; dorsal 120 m at $r = 1,019$ to 1,219, $z = -150$ to -30 , above the gallery spurs and below the mount recesses.
Cavity screen emitters	Eight apertures of 90 m on the knuckle at $r = 1,097$, $z = -500$, clocked to the Lance longitudes from 22.5° , over generator halls of 120 m at $r = 1,190$ to 1,310, $z = -290$ to -170 (Section 6.1).
Drive emitter	Sixteen segments across the fairing's aft boundary, each about 169 m of face width over the full 630 m section, the face standing 20 m proud of the structural boundary.

B.9 Collar, cap and flag stack

- Sixteen viewport bays on a 137 m pitch at mid-radius, eight clocked to the web longitudes and eight between. Glazing 100 m by 18 m a bay in panes of 10 by 6 m, armored transparisteel behind quadanium shutters; piers 37 m wide. Collar structural depth 8 m, crown at +100, travel 45 m to a stowed crown flush with the dorsal curve.
- Flag plot is an annular gallery at $r = 210$ to 250 on levels $n = 42$ to 46, wrapped around the central core's crown under the fixed cap, which clears the crown by 14 m: 4 m of structure and the 10 m grade. The inner radius stands it clear of Praetorian's bowl shell, whose outer surface passes $r = 205$ at the plot's lowest level and

narrows to $r = 165$ at its highest. The central core's command deck lies at $z = +28$ inside the core, one level below the plot, which is the step down Section 15 describes.

- Combat bridge in the citadel at $n = -9$, $r = 380$ to 560 on the 0° spine, a hall 180 m by 120 m with identical stations and no viewports.

B.10 Aft block

- The block spans $\theta = 135^\circ$ to 225° at full depth, from $r = 1,000$ outward. Four reactor halls stand off the web longitudes at 150° , 165° , 195° and 210° , each 300 m tangential and 498 m deep between $z = -566$ and -68 , running outboard to the fairing curve less 40 m: the outer pair from $r = 1,420$, the inner pair from $r = 1,490$, since the fairing is deeper nearer the stern. Containment spheres of 180 m sit at $r = 1,570$ on the outer pair and $r = 1,640$ on the inner, both at $z = -317$.
- Each hall feeds four of the sixteen emitter segments directly and cross-ties to the 180° spine. Armored bulkheads of 20 m separate each hall from the next and from the ordnance blocks inboard of it, and no hangar volume lies inboard of any of them. The outer band carries no endurance plant in this arc: the volume is propulsion, ordnance and their protection (Section 14).
- The fairing's dorsal surface is flat at $z = 0$ and carries the full dorsal grade, so from above the stern reads as a widening of the disc rather than a structure on it.

B.11 Endurance plant and habitability

- Growing decks: two blocks a sector at $r = 1,350$ to $1,550$, $z = -386$ to -344 — eight primary bays subdivided at 3 m into sixteen tiers — each 184 m tangential and $36,800$ m² in plan. Five sectors carry theirs flanking the auxiliary block at 100 to 284 m either side of the spine. The two sectors flanking the aft block carry both of theirs on the bow side, at 100 to 284 and 300 to 484 m, because the aft half of their outer band is propulsion and ordnance. The aft sector carries none. That is $73,600$ m² to a sector across seven sectors, $515,200$ m² in all, against the $516,000$ m² of footprint and 8.25 million m² of growing area Section 12 sets.
- Recycling, water and decade-scale stores occupy the outer band from $n = -45$ to -26 , between $r = 1,150$ and $r = 1,500$. Nothing but weapons, their magazines and their feed stands outboard of $r = 1,500$ between $z = -142$ and -532 (Annex C). Fabrication and magazines sit in the citadel between $n = -20$ and -1 .
- Berthing for the hull's $51,780$ occupies levels $n = 20$ to 46 , at about 35 m² per person of cabin and immediate service and 110 m² of total habitable claim including messes, medical, training and recreation: some 5.7 million m². Air group handling and ready rooms sit on $n = 12$ to 19 , immediately above the upper gallery.

B.12 De-stack scheme

- The disc explodes along one axis. Eight slabs lift dorsally, the aft block draws aft, and the nine cores drop, each carrying its own sub-stack. Suggested spacing is 150 m between slabs, 600 m for the aft block and 900 m for the cores, which keeps the whole assembly inside three ship diameters.

Layer	Extent	Draw
Collar and fixed cap	$r \leq 450$, above $z = -10$	Dorsal
Dorsal shell and battery band	$n = 31$ to 37 , with the 10 m grade	Dorsal
Berthing and air group	$n = 12$ to 30	Dorsal
Upper gallery	$n = 11$, ± 12 m	Dorsal
Handling and stores	$n = 1$ to 10	Dorsal
Lower gallery	$n = 0$, ± 12 m	Dorsal
Citadel and endurance plant	$n = -45$ to -1	Dorsal
Ventral traffic and shell	$n = -65$ to -46 , with the 4 m grade	Ventral
Aft block	$\theta 135^\circ$ to 225° , $r \geq 1,000$, full depth	Aft
Cores, nine	Spheres at their stations	Ventral

- Each core sub-stacks on the same principle: northern cap with the drive cluster, upper hangar deck, berthing, lower hangar deck, equatorial box, vehicle decks, southern band, polar cap.

B.13 Model budget

- The shell as a lathe at 1,024 segments, with the fairing, bowls, wells and collar well as boolean subtractions, comes to roughly two million triangles including the nine cores — a desktop model.
- Deck plates cut to the hull profile with the cavity, wells and trunks subtracted, at 6 m pitch, run to twenty or thirty million triangles across 103 levels. This is the level at which a de-stack reads correctly, and it is the one to build first.
- Room-level interiors are a different category: 805 million m^2 of deck at any useful subdivision is 10^8 to 10^9 triangles, which is a streaming and instancing problem, not an asset. Build compartment kits and instance them, and exploit the eight-fold symmetry — six segments identical, two aft segments unique, one collar, two core types.
- Nothing in this annex requires interpretation by hand. Every surface in B.2 is a profile curve, every station in B.6 and B.8 is an angle and a radius, and the deck stack is one arithmetic series. A parametric build is also the cheapest consistency check the specification has: it will show at once whether the spurs clear the Lance wells, whether sixteen mouths and sixteen boom recesses fit the knuckle circumference, and whether the auxiliaries at $r = 1,400$ clear the ventral rings.
- The set has been checked against itself: every volume in this annex sampled against every other and against the hull envelope — 152 volumes, 271,000 points — with no clash, no envelope violation and both boom roots on the knuckle curve. Voids are tested too. Galleries and spurs are permitted to cross a bowl, because they mate with the core there, and the launch mouths are permitted to cut the knuckle, because that is what they are for; everything else that touches the envelope is an error. Re-run the check before any revision: it is the only thing in this document capable of falsifying it.

Annex C — Fit-out and services datum set

Author's reference, outside the in-universe specification. Annex B fixes the ship's structure; this annex fixes what stands inside it. Habitability figures follow the United States Navy shipboard habitability criteria, which is the only body of practice that sizes a warship's interior by population rather than by floor plan; where two editions of it disagree the governing figure is given with the alternative, as Annex A does for source material. Everything else

originating here is a decision of this revision.

C.1 Population and basis

- The complement of Section 10 — 109,592 — is the whole population aboard, the embarked Corps included. Every per-person figure in this annex keys to it. The design population is 130,000, being the complement plus the 5 to 10 percent design margin and 10 percent service-life allowance the criteria require.
- Food is the exception and keys to 183,340, the whole task force, because Section 12 sizes the growing decks for the group the flag replenishes.
- Rates that matter at this scale: 92 t of oxygen consumed and 110 t of carbon dioxide produced a day; 15 MW of metabolic heat; 16,600 m³ of fresh water a day against a stowage of the same; 62 t of ice; 1,193 t of laundry a week; 198 t of dry cleaning a month.

C.2 Clearances and circulation

Element	Datum
Headroom, walking areas	1,956 mm unobstructed, which also clears the tallest routine droid at 1,850 mm.
Passage hierarchy	Trunk thoroughfare 3,000 mm wide by the full 4,500 mm clear; primary 1,800 mm with handrails both sides; secondary 1,120 mm; berthing walkway 914 mm; equipment access 610 mm.
Doors	Personnel 914 × 2,032 mm clear; double and main-passage 1,400 × 2,000 mm; blast doors two-leaf converging, 4 m turadium, 20 × 12 m at citadel approaches, in series of three.
Hatchability	Every item of furniture and equipment passes a 660 × 1,676 mm opening with 203 mm corner radii, or is built to knock down. This governs the removal routes of C.13.
Sills and ramps	No sill above 50 mm on any route a wheeled droid uses; ramps at 1:12. Sanitary boundaries carry a 152 mm coaming, ramped on both faces.
Stairs and ladders	Stairs 20° to 50°, ladder-stairs 50° to 75°, ladders 75° to 90°; escape ladders 600 mm clear at not more than 60°.
Egress	Second means of egress from any space of 21 or more accommodations; stair and door widths 900 mm plus 10 mm for each person above 90 assigned to them.

C.3 Berthing

- The unit is a 48-person compartment: eight cubicles of six, three-high tiers, facing pairs with a 914 mm main walkway and 686 mm alongside each tier. Berth pan 2,040 × 800 mm; mattress 2,032 mm long in a tenth of tiers and 1,930 mm in the rest; 508 mm clear above the mattress; 152 mm beneath the bottom pan. One light, one receptacle, one privacy curtain, a full partition and an HVAC terminal to each berth; kick-out panels on inner lower tiers.
- Stowage to each berth: 0.37 m³ of locker, 254 mm of hanging space, 0.03 m³ of seabag stowage and 0.014 m³ for soiled linen. Across the complement that is 40,300 m³ of personal locker volume — a figure worth carrying, because it is larger than the citadel.
- Compartment count: 2,284 at 48, 18,266 cubicles at six. Officers take one- and two-person staterooms at 3.3 to 8.4 m² of unobstructed area by rank; no bunkroom exceeds six berths. Berths run fore and aft, with not more

than three in ten athwartship.

- With its sanitary module the compartment runs about 2.4 m² of deck per person, so all berthing aboard is roughly 263,000 m² — a third of one percent of the deck the ship has. Berthing is never volume-constrained on this hull; it is constrained by the adjacency rules of C.4.

C.4 Sanitary

- Fixture ratios, governing figures from the current criteria with the older limits as the alternative: one washbasin to 12 (alternative 20), one water closet to 14 (28), one shower to 25 (30). Across the complement that is 9,133 washbasins, 7,828 water closets and 4,384 showers; on the older limits, 5,480, 3,914 and 3,653. Officers and senior ratings are served more generously and are additional.
- Fixture geometry: washbasins on 610 mm centres in continuous countertop with a full-length segmented mirror, a shelf and two receptacles each; urinals on 686 mm centres with partitions; water closets in cubicles of 762 × 762 mm with a privacy door; showers in individual stalls of 762 × 762 mm grouped about a common drying area behind a joiner bulkhead.
- Three adjacency rules bind harder than the counts, and they are what make the berthing module a module: the sanitary space serving a berthing compartment stands on the same deck as it; no one crosses a primary passage or a public area to reach it; and washroom access never passes through the water closet area. A 48-person compartment therefore carries its own four washbasins, four water closets and two showers, in 22 m² off its own lobby.
- Sanitary spaces take 15 air changes an hour with an exhaust terminal at each shower group and each closet group, are held at 24 °C, and drain to a coamed deck. Water heaters stand outside them.

C.5 Messing and food service

- Mess seats at 30 percent of accommodations — 32,878 — on 533 mm of table edge each for crew and 610 mm for officers and senior ratings. The carrier figure of 14 percent is the alternative and assumes continuous flight operations across more sittings; at this complement it would give 15,343.
- Messroom clearances: primary passages 914 mm, secondary 762 mm, 610 mm back to back between seated diners. Every messroom is designed to double as a training and recreation space, which is where a third of the recreation requirement in C.6 is actually met.
- Galleys, bakeries, vegetable preparation, sculleries and utensil wash are sited one to a sector with the mess they serve. No sewage or toxic line passes above a food preparation space. Serving lines are lit to 450 lux, messrooms to 300.
- Stores: dry, chill, freeze and thaw rooms sized to the standing endurance, drawn from the growing decks and the fabrication plant rather than from bunkering. Ice plant at 62 t a day. Solid-waste processing sits with the galley, not remote from it.

C.6 Personal services and community

Service	Provision
Recreation and lounge	Space for a third of the complement at once — 36,531 — exclusive of messrooms; one writing surface to ten and three seats to ten, so 10,959 surfaces and 32,878 seats.
Physical fitness	Three 30-minute periods a person a week against 112 hours of weekly availability; 1,468 equipment stations, with a dedicated facility in every sector and every core.
Library	Separate, with shelving, tables and chairs; one to a sector.
Chapel	Separate space above 1,000 accommodations; here one to each sector and each core, seventeen in all, at 45 seats.
Barber	One chair to each 300 accommodations: 366 chairs.
Ship store, post office	Walk-in stores with over-the-counter outlets at sector level; a secure post office with parcel stowage, and dispatch handling for the courier flight.
Snack bar and vending	Required above 500 accommodations; one snack bar a sector in a high-traffic node away from the messrooms, with separate vending areas.
Laundry	1,193 t a week over a 96-hour laundry week, finished work at 5.4 kg a week for each officer and senior rating and 3.2 kg for each crew; dry cleaning at 1.8 kg an accommodation a month. Irons and boards in every berthing space.

C.7 Medical

- Beds at five to the thousand, 548 in all, distributed across the nine cores and the hull with primary medical in Praetorian (Section 3.4). Surge capacity for mass casualty is additional and takes the adjacent messrooms, which is why they are sited near the galleries.
- Bacta tanks: 1.5 × 1.5 m in plan and 3.0 m overall including top service, 1,100 kg full, one patient each, consuming 1 L an hour with fill, drain and filtration plant behind. Tank stations are grouped in fours off each surgical suite.
- Operating theatres at 55 m², 20 air changes an hour, positive pressure, sheathed for cleanability; isolation suites at 12 changes and negative pressure. Battle dressing stations one to each gallery arc and each core, stocked with portable medical lockers, litters and oxygen.

C.8 Environment

- Six air changes an hour in every manned space, fifteen in sanitary spaces, twenty in theatres; 2.4 L/s of replenishment air a person, or 260 m³/s across the ship before recirculation. A 100 m² compartment therefore takes a branch duct of about 500 × 500 mm, which is what sets the depth of the services band in C.9.
- Temperatures: not below 18 °C in living, working and control spaces, 24 °C in sanitary and medical, not above 26 °C dry bulb anywhere manned, to a tolerance of one degree.
- Airborne noise limits by category: 60 dBA where speech must carry twelve feet — bridges, combat bridge, control rooms, offices, training spaces; 65 in medical, library and chapel; 70 in berthing, lounges, messrooms and barber shops; 84 in machinery, laundry and workshops. These set acoustic treatment thickness, which is volume in the services band.
- Illuminance: passageways 75 lux; berthing and staterooms 75; living spaces and sanitary 150; messing, food service, library, ship store and post office 300; writing surfaces, serving lines, ranges and barber chairs 450; shower areas 32. Maximum to minimum between fixtures not worse than two to one, task to surround not worse than ten to one.

C.9 The services band

- Of the 6 m deck pitch, 4.5 m is clear and 1.5 m is structure and services. That 1.5 m is stacked, top down: 300 mm of deck plate and girder, 400 mm of ventilation, 300 mm of power and data, 300 mm of fluids, 200 mm free for pulls and future work. Nothing crosses a zone except at a stated riser.

Service	Datum
Ventilation	Branch 500 × 500 mm a compartment; deck mains 1,200 × 800 mm; risers on the web longitudes; velocities 3 m/s in branches, 8 in mains.
Power	Ladder tray 600 mm wide at 40 percent fill on the outboard side of the band; feeders from the sector load centre, one a compartment; bend radius twelve diameters.
Data and control	Tray 300 mm on the inboard side, 300 mm clear of power, with fire and damage-control circuits in their own tray and route.
Potable and grey water	Riser 150 mm on the spine, branch 80 mm, velocity not above 2.4 m/s; hot water at every basin and shower within ten seconds.
Waste	Gravity branch 200 mm sloped to a sector collection main of 400 mm; no waste line above a food space.
Coolant	Sector loop 300 mm flow and return, isolable at each compartment; no coolant line over a berth.
Firemain	200 mm ring on each deck, cross-connected at every web longitude, with a hose station at every stair enclosure and every 40 m of primary passage.

C.10 Furniture and fittings schedule

Item	Datum
Berth	Pan 2,040 × 800 × 770 mm, three-high tier 2,130 mm overall above the 152 mm underclearance.
Locker, crew	0.37 m ³ , 1,800 × 400 × 520 mm, in the cubicle end or beneath the bottom berth.
Writing surface and chair	900 × 600 mm at 740 mm, one to ten accommodations.
Mess table	1,800 × 750 mm at 740 mm, seating four at 533 mm of edge.
Seated console	Work surface 1,800 × 900 mm at 740 mm; knee space 510 wide, 640 high, 460 deep; primary display within 15° of the line of sight; secondary displays within 35°.
Standing console	Work surface at 1,000 mm; controls between 865 and 1,780 mm, preferred 1,020 to 1,420.
Controls	Push-buttons 19 mm square on 25 mm centres; keypads 10 to 19 mm keys; emergency controls guarded and 50 mm clear of any other control.
Bulkhead display	1,200 × 700 mm at 1,400 mm to centre, one at every compartment entrance carrying compartment number, muster station and damage-control state.
Viewport, collar	Panes 10 × 6 m in armored transparisteel, sixteen bays of 100 × 18 m, piers 37 m, quadanium shutters behind.
Door furniture	Lever handles at 1,000 mm; keypads at 1,200 mm; dogging levers at 1,100 mm; all within the standing control band.
Damage-control locker	2,400 × 1,000 × 2,100 mm, one at every repair station and every 80 m of primary passage.

C.11 Droids

- Three envelopes govern: bipedal service droids at 1,850 mm tall and 700 mm wide; astromechs at 1,200 mm tall and 800 mm wide on three legs; mouse droids at 600 × 540 × 400 mm. Everything in C.2 clears all three, which is the point of setting sills at 50 mm and door widths at 914.
- Mouse-droid service ducts 600 × 400 mm run the full deck stack alongside the data tray, with grilles into every compartment and a charging niche at every sector node. Astromech ports are fitted at the flight-operations consoles, the combat bridge and each core command deck.
- Droid maintenance and charging bays, one a sector and one a core, with parts stowage; power droids circulate the same routes as personnel and are not given separate ones.

C.12 Weapon installations

- Every mount is a stack, not an aperture: shutter, mount and its stroke, barbette, feed trunk, ready magazine, thermal sink. Annex B gives the first of those; the rest is here, and it is why the outer 178 m of the hull carries nothing else.

Installation	Heavy, rim	Medium, faces	Close-in
Aperture	30 m	15 m	8 m
Barbette	44 m	22 m	12 m
Recess depth	30 m	45 m	20 m
Deployed stroke	12 m	8 m	4 m
Swept envelope	Hemisphere of 25 m about the trunnion	14 m	8 m
Feed trunk	15 m over 90 m	8 m over 40 m	Ready rounds at the mount
Ready magazine	40 × 40 × 30 m	20 × 20 × 15 m	None
Thermal sink	30 × 30 × 20 m	12 × 12 × 10 m	Mount mass only
Count	591	117 / 234 / 261	1,280 hull, 640 cavity

- No magazine, trunk, hall or stores volume enters a mount's swept envelope. The envelope is a keep-out in the deployed state only, but it is modelled permanently, because a mount that cannot traverse is a mount that is not there.
- Installed volume, bottom up: 53 million m³ of mounts, barbettes, trunks, magazines and sinks, plus 71 million m³ of ordnance blocks aft — 124 million m³, or 2.6 percent of the hull.

C.13 Safety, escape and damage control

- Escape pods at twelve persons, one to each berthing compartment and each watch station group: 9,133 pods, in bays on the rim and the ventral annulus with launch rails clear of the battery band. Rations for two weeks, beacon, and no droids aboard.
- Damage-control lockers and repair stations as C.10; hose stations at every stair enclosure and every 40 m of primary passage; portable extinguishers at every compartment entrance; fixed suppression in machinery, magazine, galley and workshop volumes.
- An emergency breathing device over every berth and at every watch station; emergency air stations at every sector node; low-location photoluminescent egress marking on every route, with muster stations at the gallery arcs and embarkation at the pod bays.
- Equipment removal routes from every machinery block to a launch mouth or a gallery, sized to the largest single item in that block and marked on the deck. This is the rule that most often fails late, and it is checked in the model rather than asserted here.

C.14 Volume budget

- The enclosed volume is 4.83 billion m³. Annex B allocates the machinery; this annex allocates the rest, in priority order, so that no later revision can quietly spend volume twice.

Claim	Share	Volume
Structure: plating, framing, the 1.5 m services band	25%	1,210 M m ³
Compartmented void, as armour depth and damage volume	15%	725
Feedstock, raw material and structural spares	10%	480

Claim	Share	Volume
Task-force replenishment inventory	10%	480
Hypermatter bunkering and reactant handling	8%	385
Machinery: reactor, auxiliary, shield and screen halls	6%	290
Occupation stores and garrison modules	5%	240
Industrial: foundry, fabrication, repair, salvage	5%	240
Endurance plant: agriculture, recycling, water, atmosphere	4%	195
Air group, galleries, mouths, vehicle and cargo handling	3%	145
Access outside the services band: trunks, shafts, passages	3%	145
Weapons: mounts, barbettes, magazines, ordnance	2.6%	124
Growth margin	2%	95
Hotel: berthing, messing, medical, recreation	1%	50

- Two entries deserve their place rather than being residue. Void is armour: Section 1 builds this hull to absorb damage rather than avoid it, and compartmented void behind the rim plating is what absorbs it — naming it stops a later revision recovering it for stores. Replenishment inventory is a fleet train carried inside one hull, because Section 18.4 supplies forty-six escorts indefinitely from these holds.
- The hotel line is the smallest claim on the ship and the one every reader expects to be largest. That is the honest shape of a vessel whose complement is a hundred thousand and whose volume is four and a half billion cubic metres.

Annex D — Finish, lighting and markings

Author's reference, outside the in-universe specification. Annex B fixes what the ship is; Annex C fixes what stands inside it; this annex fixes what both look like. Almost all of it follows from facts the specification already states — who builds each part (Section 19), what each part is made of (Section 16), who serves in it (Section 10), and that the hull never enters an atmosphere (Section 2.6). Where the derivation runs out, the annex says so and leaves the space deliberately open.

D.1 Finish follows the builder

- Five yards build this ship and none of them finishes work the same way. The zones are legible from the consortium table and should be modelled as such: the hull, its framing and its exterior are Kuat work; the nine cores, their ground complement and their vehicle decks are Rothana; the endurance plant, the recycling and water spaces and the damage-control fit are Mon Calamari; the galleries, mouths and air group spaces are Kuat Systems Engineering; the hyperdrive spaces are Hoersch-Kessel.
- Kuat practice is flat, machined and anonymous: surfaces ground true, seams closed, fastenings concealed, no ornament and no contrast. Rothana practice is equipment-grade and openly utilitarian: exposed fixings, heavy wear plate, hard-edged colour on anything that moves or is handled. Mon Calamari practice is the opposite of both, curved where it can be, sealed and coved at every junction, colour-coded by fluid and system, and noticeably better lit. Kuat Systems marks everything it builds with handling lines, bay numerals and hazard striping because its spaces are worked by crews under time pressure.

- A viewer who walks from a gallery into a core and then into the growing decks should be able to tell, without being told, that three different firms built them.

D.2 Exterior finish

Surface	Finish
Dorsal face	Neutronium-impregnated quadanium, 10 m grade: dense mid-grey with a faint blue cast, matte and deliberately non-specular — 8.86 million m ² of bright plate would be visible across a system. Panel modules on the 20 m frame grid, seams closed and ground, no fastenings visible.
Rim	Same plate at 4 m, reading as a repeating 40 m module of shutter and recess over three rows, with the shield ring apertures interrupting it at five forward longitudes.
Ventral annulus and cavity	The only busy exterior on the ship: sixteen mouths, sixteen boom recesses, eight screen emitters, nine wells, the crown and the knuckle, all marked and all lit.
Drive fairing and emitter	Dallorian throat liners, heat-discoloured from bronze through straw to blue-grey across the sixteen segments in service; the fairing crown darkened for two hundred metres forward of the face.
Bowls and wells	Alusteel over ferrocabon, lighter and satin against the hull plate, with duranium keys and dogs left bright and visibly worn at the bearing faces.
Core southern cap	Quadanium substrate with refractory ceramic facing: pale, chalky, and scorched to grey-brown at the stagnation region after any landing. This is the one surface aboard that shows a core's history.
Weathering	Micrometeorite pitting at roughly one pit above 1 mm per square metre per decade, concentrated on the dorsal face and the rim's leading arc; no corrosion anywhere, since the hull never meets an atmosphere; repair plate left unblended and visibly newer.

D.3 Exterior markings

- Hull name and number on the dorsal face at two-thirds radius, in low-contrast dark-on-grey at a size legible from a ship's length: the name forward, the division numeral aft. The class badge sits on the fixed cap, which is the one part of the dorsal surface nothing crosses.
- Sector numerals one to eight at the rim on each web longitude, repeated inside the cavity at each boom station, so that formation, docking and boom work all reference the same eight marks.
- Launch mouths numbered one to sixteen with approach chevrons on the knuckle either side; wells lettered for their cores; every aperture on the ship ringed in hazard marking that is invisible at range and unmistakable at fifty metres.
- Core designations — Praetorian, Lance One through Lance Eight — painted large on each southern cap, which is the face that is uppermost when a core is landed and the only marking on this ship intended to be read from the ground.

D.4 Exterior lighting

- Cruising: position lights at the eight rim longitudes, white over the forward 270° and red over the drive arc; a beacon at the collar crown while it is extended; the cavity floodlit amber whenever the mouths are open.

- Flight operations: approach lighting framing each mouth in a sequenced ladder, with the two mouths of a segment coded left and right; boom stations lit green when stowed and amber when deployed; the holding station beneath the cavity marked by four datum lights on the dish rim.
- Action: all exterior lighting extinguished, including the cavity. The ship that presents no aim point at the hub does not carry one in light either.

D.5 Interior finish by zone

Zone	Finish
Citadel	Permacite bulkheads, pale and seamless, unpainted; turadium blast doors near-black and matte; no sheathing, no carpet, nothing applied that could burn. The most austere space aboard.
Galleries and mouths	Alusteel joiner, satin light grey, with Kuat Systems deck marking: handling lanes, bay numerals, hold-short lines, hazard striping at every lift edge and mouth sill.
Berthing and messing	Alusteel joiner with a sector colour band at 1,400 mm on every bulkhead; deck covering in a mid tone that shows neither dust nor wear quickly; sheathing only where the criteria allow it.
Sanitary, galley, medical	Fully sheathed and coved for cleaning, gloss finish, white to pale green; the only high-reflectance surfaces on the ship.
Endurance plant	Mon Calamari work: sealed wet spaces, coved junctions, fluid systems colour-coded by service, growing decks under magenta-white horticultural light that spills into the access lanes.
Machinery and weapons	Bare doonium and ferrocargon, agrinium underlayer visible in reactor spaces as a dull green-grey, dallorian bearing faces bright; walkways in open grating; everything labelled and nothing decorated.
Cores	Rothana work: heavy wear plate on vehicle decks, hard-edged yellow and black at ramps and gear bays, unit markings applied by the embarked regiments and not by the yard.

D.6 Wayfinding and numbering

- The hull is eight-fold symmetric and a person standing in it has no natural way to know which sector they are in. Every sector therefore carries a colour and a numeral, banded at 1,400 mm on every bulkhead and repeated at every junction; the bands run continuously so that following one arrives at that sector's spine.
- Compartment numbering is four-part: sector, frame ring, deck level, usage. Ring numbers run one to eighty-four outward from the axis on the 20 m frame spacing; levels use the deck grid n of Annex B; usage letters are L living, M messing, A stores, E engineering, W weapons, C control, F flight, S sanitary, H medical. A compartment reads 4-52-N06-L, which locates it to twenty metres anywhere in four and a half billion cubic metres.
- Every compartment entrance carries the display of C.10: number, muster station, damage-control state, and the sector colour.

D.7 Interior lighting

- General lighting is solid-state, 4,000 K in working and living spaces, 5,000 K in medical and inspection spaces, 3,000 K in lounges, messrooms and chapels — the one aesthetic concession the ship makes, and the only warm light aboard outside the growing decks.

- Levels as C.8. Fixtures on 4 to 6 m centres at a 4.2 m mounting height, coordinated with the ventilation terminals so that neither is ever directly over a berth.
- Emergency lighting at every junction, stair and door, with photoluminescent low-location marking on every escape route that reads with no power at all. Action lighting reduces general illumination by half and shifts control spaces to red.

D.8 Moving parts and their states

- A model that never moves is a model that hides its mistakes. Every closure aboard has two states and a stated travel: iris shutters of sixteen leaves over the collar well, 45 m of collar travel to a crown at +100; mount strokes of 12, 8 and 4 m by class; two-part closures at every thruster well, hangar mouth, boom recess and pad bay, outer shutter flush and inner seal deeper; core ventral ramps of 40 × 15 m dropping to a gear-borne deck; gear legs of 50 to 105 m by latitude.
- The cavity screen has four states — off, attenuating at the rim plane, hardened 10 m below it, and drawn inboard against the dish — and they are visually distinct, because everything about how the ship is read from outside depends on which one it is in.

D.9 What is deliberately not specified

- Everything above is the fixed palette. Inboard of the joiner line, and only there, the following are deliberately left open and should be generated rather than specified: personal effects in berth cubicles, curtains and holos and the small disorder of a hundred thousand people; art and unit insignia in messrooms, lounges and chapels; regimental markings on core vehicle decks and air group craft; wear paths on deck coverings at high-traffic nodes; and mismatched replacement panels wherever damage has been made good.
- Two rules bound the randomness. Nothing personal appears on the exterior, in the citadel, or in any machinery or weapons space. And nothing generated may obstruct a marked route, a removal path, a damage-control fitting or an escape marking — the lived-in look stops precisely where the safety case begins.
- The clone technical branches serve the hull for its operational life while the commissioned chain rotates. The interior should show that: long-tenured spaces — engineering, damage control, hull maintenance — accumulate the most personalisation, and officer country the least.